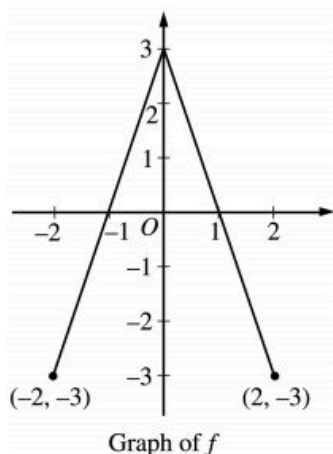


5-6b

1.



The graph of the function f shown above consists of two line segments. Let g be the function given by $g(x) = \int_0^x f(t) dt$.

- Find $g(-1)$, $g'(-1)$, and $g''(-1)$.
- For what values of x in the open interval $(-2, 2)$ is g increasing? Explain your reasoning.
- For what values of x in the open interval $(-2, 2)$ is the graph of g concave down? Explain your reasoning.
- On the axes provided, sketch the graph of g on the closed interval $[-2, 2]$.

Part A

1 point is earned for $g(-1)$

1 point is earned for $g'(-1)$

1 point is earned for $g''(-1)$

$$g(-1) = \int_0^{-1} f(t) dt = - \int_{-1}^0 f(t) dt = -\frac{3}{2}$$

$$g'(-1) = f(-1) = 0$$

$$g''(-1) = f'(-1) = 3$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g(-1)$

1 point is earned for $g'(-1)$

5-6b

1 point is earned for $g''(-1)$

$$g(-1) = \int_0^{-1} f(t)dt = - \int_{-1}^0 f(t)dt = -\frac{3}{2}$$

$$g'(-1) = f(-1) = 0$$

$$g''(-1) = f'(-1) = 3$$

Part B

1 point is earned for the interval

1 point is earned for the answer

g is increasing on $-1 < x < 1$ because $g'(x) = f(x) > 0$ on this interval.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the interval

1 point is earned for the answer

g is increasing on $-1 < x < 1$ because $g'(x) = f(x) > 0$ on this interval.

Part C

1 point is earned for the interval

1 point is earned for the reason

The graph of g is concave down on $0 < x < 2$ because $g''(x) = f'(x) < 0$ on this interval.

or

because $g'(x) = f(x)$ is decreasing on this interval.



0	1	2
---	---	---

The student response earns all of the following points:

5-6b

1 point is earned for the interval

1 point is earned for the reason

The graph of g is concave down on $0 < x < 2$ because $g''(x) = f'(x) < 0$ on this interval.

or

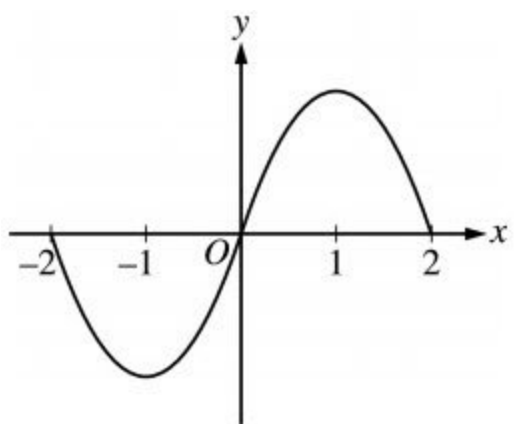
because $g'(x) = f(x)$ is decreasing on this interval.

Part D

1 point is earned for $g(-2) = g(0) = g(2) = 0$

1 point is earned for appropriate increasing/decreasing and concavity behavior

<-1> vertical asymptote



0	1	2
---	---	---

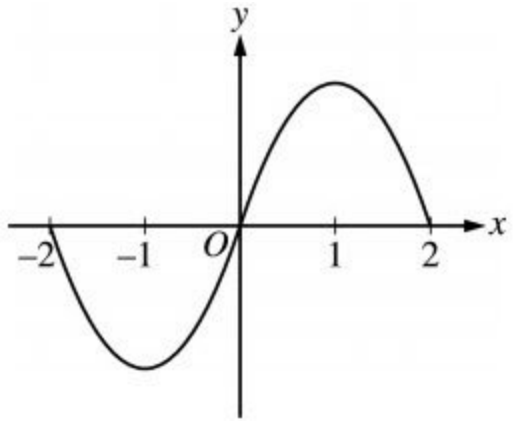
The student response earns all of the following points:

1 point is earned for $g(-2) = g(0) = g(2) = 0$

1 point is earned for appropriate increasing/decreasing and concavity behavior

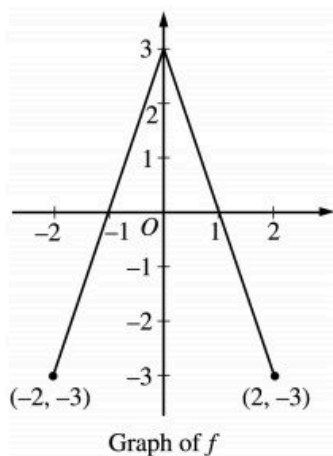
<-1> vertical asymptote

5-6b



5-6b

2.



The graph of the function f shown above consists of two line segments. Let g be the function given by $g(x) = \int_0^x f(t) dt$.

For what values of x in the open interval $(-2, 2)$ is the graph of g concave down? Explain your reasoning.

Part C

1 point is earned for the interval

1 point is earned for the reason

The graph of g is concave down on $0 < x < 2$ because $g''(x) = f'(x) < 0$ on this interval.

or because $g'(x) = f(x)$ is decreasing on this interval.



0	1	2
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The student response earns all of the following points:

1 point is earned for the interval

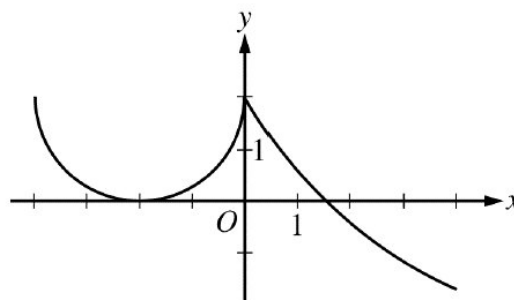
1 point is earned for the reason

The graph of g is concave down on $0 < x < 2$ because $g''(x) = f'(x) < 0$ on this interval.

or because $g'(x) = f(x)$ is decreasing on this interval.

5-6b

3.

Graph of f'

The derivative of a function f is defined by $f'(x) = \begin{cases} g(x) & \text{for } -4 \leq x \leq 0 \\ 5e^{-x/3} - 3 & \text{for } 0 < x \leq 4 \end{cases}$.

The graph of the continuous function f , shown in the figure above, has x -intercepts at $x = -2$ and $x = 3 \ln\left(\frac{5}{3}\right)$. The graph of g on $-4 \leq x \leq 0$ is a semicircle, and $f(0) = 5$.

(a) For $-4 < x < 4$, find all values of x at which the graph of f has a point of inflection. Justify your answer.

(b) Find $f(-4)$ and $f(4)$.

(c) For $-4 \leq x \leq 4$, find the value of x at which f has an absolute maximum. Justify your answer.

Part A

1 point is earned if identifies $x = -2$ or $x = 0$

1 point is earned for answer with justification

f' changes from decreasing to increasing at $x = -2$ and from increasing to decreasing at $x = 0$. Therefore, the graph of f has points of inflections at $x = -2$ and $x = 0$.



0	1	2
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The student response earns all of the following points:

1 point is earned if identifies $x = -2$ or $x = 0$

1 point is earned for answer with justification

f' changes from decreasing to increasing at $x = -2$ and from increasing to decreasing at $x = 0$. Therefore, the graph of f has points of inflections at $x = -2$ and $x = 0$.

Part B

1 point is earned for integral of $f(-4)$

5-6b

1 point is earned for value of $f(-4)$

1 point is earned for integral of $f(4)$

1 point is earned for antiderivative of $f(4)$

1 point is earned for value of $f(4)$

$$\begin{aligned} f(-4) &= 5 + \int_0^{-4} g(x) dx \\ &= 5 - (8 - 2\pi) = 2\pi - 3 \end{aligned}$$

$$\begin{aligned} f(4) &= 5 + \int_0^4 (5e^{-x/3} - 3) dx \\ &= 5 + (-15e^{-x/3} - 3x) \Big|_{x=0}^{x=4} \\ &= 8 - 15e^{-4/3} \end{aligned}$$



0	1	2	3	4	5
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The student response earns all of the following points:

1 point is earned for integral of $f(-4)$

1 point is earned for value of $f(-4)$

1 point is earned for integral of $f(4)$

1 point is earned for antiderivative of $f(4)$

1 point is earned for value of $f(4)$

$$\begin{aligned} f(-4) &= 5 + \int_0^{-4} g(x) dx \\ &= 5 - (8 - 2\pi) = 2\pi - 3 \end{aligned}$$

$$\begin{aligned} f(4) &= 5 + \int_0^4 (5e^{-x/3} - 3) dx \\ &= 5 + (-15e^{-x/3} - 3x) \Big|_{x=0}^{x=4} \\ &= 8 - 15e^{-4/3} \end{aligned}$$

5-6b

Part C

1 point is earned for the answer

1 point is earned for the justification

Since $f'(x) > 0$ on the intervals $-4 < x < -2$ and $-2 < x < 3 \ln\left(\frac{5}{3}\right)$, f is increasing on the interval $-4 \leq x \leq 3 \ln\left(\frac{5}{3}\right)$.

$3 \ln\left(\frac{5}{3}\right)$.

Since $f'(x) < 0$ on the interval $3 \ln\left(\frac{5}{3}\right) < x < 4$, f is decreasing on the interval $3 \ln\left(\frac{5}{3}\right) \leq x \leq 4$.

Therefore, f has an absolute maximum at $x = 3 \ln\left(\frac{5}{3}\right)$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

Since $f'(x) > 0$ on the intervals $-4 < x < -2$ and $-2 < x < 3 \ln\left(\frac{5}{3}\right)$, f is increasing on the interval $-4 \leq x \leq 3 \ln\left(\frac{5}{3}\right)$.

$3 \ln\left(\frac{5}{3}\right)$.

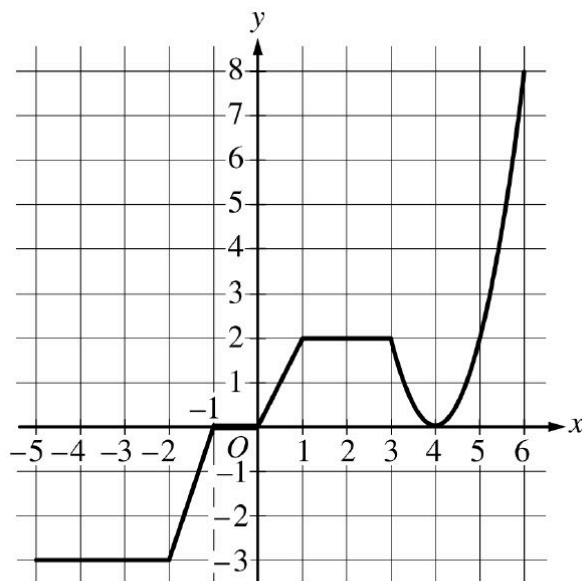
Since $f'(x) < 0$ on the interval $3 \ln\left(\frac{5}{3}\right) < x < 4$, f is decreasing on the interval $3 \ln\left(\frac{5}{3}\right) \leq x \leq 4$.

5-6b

Therefore, f has an absolute maximum at $x = 3 \ln \left(\frac{5}{3} \right)$.

5-6b

4.

Graph of g

The graph of the continuous function g , the derivative of the function f , is shown above. The function g is piecewise linear for $-5 \leq x < 3$, and $g(x) = 2(x - 4)^2$ for $3 \leq x \leq 6$.

(a) If $f(1) = 3$, what is the value of $f(-5)$?

(b) Evaluate $\int_1^6 g(x) dx$.

(c) For $-5 < x < 6$, on what open intervals, if any, is the graph of f both increasing and concave up? Give a reason for your answer.

(d) Find the x -coordinate of each point of inflection of the graph of f . Give a reason for your answer.

Part A

1 point is earned for integral

1 point is earned for answer

$$\begin{aligned}
 f(-5) &= f(1) + \int_1^{-5} g(x) dx = f(1) - \int_{-5}^1 g(x) dx \\
 &= 3 - \left(-9 - \frac{3}{2} + 1\right) = 3 - \left(-\frac{19}{2}\right) = \frac{25}{2}
 \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

5-6b

1 point is earned for answer

$$\begin{aligned} f(-5) &= f(1) + \int_1^{-5} g(x)dx = f(1) - \int_{-5}^1 g(x)dx \\ &= 3 - \left(-9 - \frac{3}{2} + 1\right) = 3 - \left(-\frac{19}{2}\right) = \frac{25}{2} \end{aligned}$$

Part B

1 point is earned for split at $x = 3$

1 point is earned for antiderivative $2(x - 4)^2$

1 point is earned for answer

$$\begin{aligned} \int_1^6 g(x)dx &= \int_1^3 g(x)dx + \int_3^6 g(x)dx \\ &= \int_1^3 2dx + \int_3^6 2(x - 4)^2dx \\ &= 4 + \left[\frac{2}{3}(x - 4)^3\right]_{x=3}^{x=6} = 4 + \frac{16}{3} - \left(-\frac{2}{3}\right) = 10 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for split at $x = 3$

1 point is earned for antiderivative $2(x - 4)^2$

1 point is earned for answer

$$\begin{aligned} \int_1^6 g(x)dx &= \int_1^3 g(x)dx + \int_3^6 g(x)dx \\ &= \int_1^3 2dx + \int_3^6 2(x - 4)^2dx \\ &= 4 + \left[\frac{2}{3}(x - 4)^3\right]_{x=3}^{x=6} = 4 + \frac{16}{3} - \left(-\frac{2}{3}\right) = 10 \end{aligned}$$

5-6b

Part C

1 point is earned for intervals

1 point is earned for reason

The graph of f is increasing and concave up on $0 < x < 1$ and $4 < x < 6$ because $f'(x) = g(x) > 0$ and $f'(x) = g(x)$ is increasing on those intervals.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for intervals

1 point is earned for reason

The graph of f is increasing and concave up on $0 < x < 1$ and $4 < x < 6$ because $f'(x) = g(x) > 0$ and $f'(x) = g(x)$ is increasing on those intervals.

Part D

1 point is earned for answer

1 point is earned for reason

The graph of f has a point of inflection at $x = 4$ because $f'(x) = g(x)$ changes from decreasing to increasing at $x = 4$.



0	1	2
---	---	---

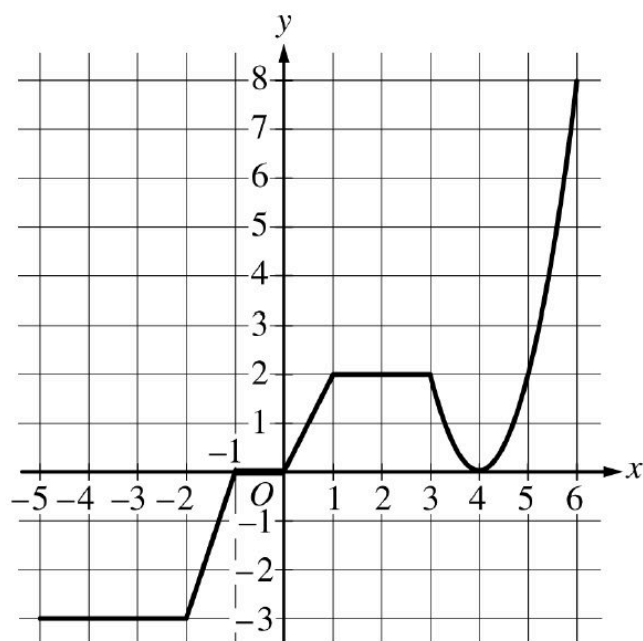
The student response earns all of the following points:

1 point is earned for answer

1 point is earned for reason

The graph of f has a point of inflection at $x = 4$ because $f'(x) = g(x)$ changes from decreasing to increasing at $x = 4$.

5-6b

Graph of g

The graph of the continuous function g , the derivative of the function f , is shown above. The function g is piecewise linear for $-5 \leq x < 3$, and $g(x) = 2(x - 4)^2$ for $3 \leq x \leq 6$.

5. For $-5 < x < 6$, on what open intervals, if any, is the graph of f both increasing and concave up? Give a reason for your answer.

Part 3C

1 point is earned for: intervals

1 point is earned for: reason

The graph of f is increasing and concave up on $0 < x < 1$ and $4 < x < 6$ because $f'(x) = g(x) > 0$ and $f''(x) = g'(x)$ is increasing on those intervals.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for: intervals

1 point is earned for: reason

The graph of f is increasing and concave up on $0 < x < 1$ and $4 < x < 6$ because $f'(x) = g(x) > 0$ and $f''(x) = g'(x)$ is increasing on those intervals.

5-6b

6. Find the x -coordinate of each point of inflection of the graph of f . Give a reason for your answer.
-

Part D

1 point is earned for: answer

1 point is earned for: reason

The graph of f has a point of inflection at $x = 4$ because $f'(x) = g(x)$ changes from decreasing to increasing at $x = 4$.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for: answer

1 point is earned for: reason

The graph of f has a point of inflection at $x = 4$ because $f'(x) = g(x)$ changes from decreasing to increasing at $x = 4$.

5-6b

7. Let h be a function defined for all $x \neq 0$ such that $h(4) = -3$ and the derivative of h is given by $h'(x) = \frac{x^2 - 2}{x}$ for all $x \neq 0$.
- (a) Find all values of x for which the graph of h has a horizontal tangent, and determine whether h has a local maximum, a local minimum, or neither at each of these values. Justify your answers.
- (b) On what intervals, if any, is the graph of h concave up? Justify your answer.
- (c) Write an equation for the line tangent to the graph of h at $x = 4$.
- (d) Does the line tangent to the graph of h at $x = 4$ lie above or below the graph of h for $x > 4$? Why?

Part A

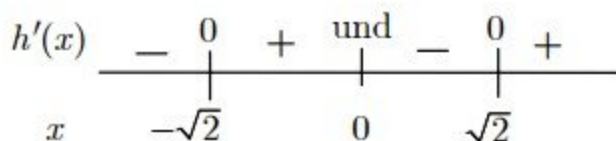
1 point is earned for: $x = \pm\sqrt{2}$

1 point is earned for the analysis

2 points are earned for the conclusions

<-1> not dealing with discontinuity at 0

$$h'(x) = 0 \text{ at } x = \pm\sqrt{2}$$



Local minima at $x = -\sqrt{2}$ and at $x = \sqrt{2}$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for: $x = \pm\sqrt{2}$

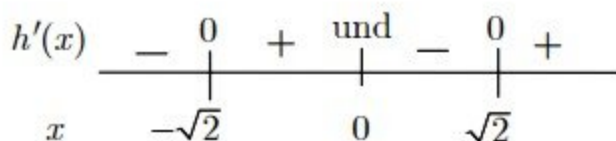
1 point is earned for the analysis

2 points are earned for the conclusions

5-6b

<-1> not dealing with discontinuity at 0

$$h'(x) = 0 \text{ at } x = \pm\sqrt{2}$$



Local minima at $x = -\sqrt{2}$ and at $x = \sqrt{2}$

Part B

1 point is earned for: $h''(x)$

1 point is earned for: $h''(x) > 0$

1 point is earned for the answer

$h''(x) = 1 + \frac{2}{x^2} > 0$ for all $x \neq 0$. Therefore, the graph of h is concave up for all $x \neq 0$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $h''(x)$

1 point is earned for: $h''(x) > 0$

1 point is earned for the answer

$h''(x) = 1 + \frac{2}{x^2} > 0$ for all $x \neq 0$. Therefore, the graph of h is concave up for all $x \neq 0$.

Part C

1 point is earned for the tangent line equation

5-6b

$$h'(4) = \frac{16-2}{4} = \frac{7}{2}$$

$$y + 3 = \frac{7}{2}(x - 4)$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the tangent line equation

$$h'(4) = \frac{16-2}{4} = \frac{7}{2}$$

$$y + 3 = \frac{7}{2}(x - 4)$$

Part D

1 point is earned for the answer with reason

The tangent line is below the graph because the graph of h is concave up for $x > 4$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer with reason

The tangent line is below the graph because the graph of h is concave up for $x > 4$.

5-6b

8. Let h be a function defined for all $x \neq 0$ such that $h(4) = -3$ and the derivative of h is given by $h'(x) = \frac{x^2 - 2}{x}$ for all $x \neq 0$.

On what intervals, if any, is the graph of h concave up? Justify your answer.

Part B

1 point is earned for: $h''(x)$

1 point is earned for: $h''(x) > 0$

1 point is earned for the answer

$h''(x) = 1 + \frac{2}{x^2} > 0$ for all $x \neq 0$. Therefore the graph of h is concave up for all $x \neq 0$.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for: $h''(x)$

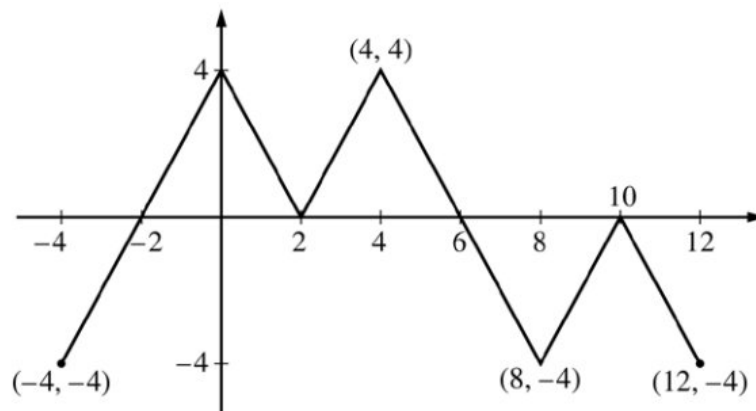
1 point is earned for: $h''(x) > 0$

1 point is earned for the answer

$h''(x) = 1 + \frac{2}{x^2} > 0$ for all $x \neq 0$. Therefore the graph of h is concave up for all $x \neq 0$.

5-6b

9.

Graph of f

The figure above shows the graph of the piecewise-linear function f . For $-4 \leq x \leq 12$, the function g is defined by $g(x) = \int_2^x f(t) dt$.

- Does g have a relative minimum, a relative maximum, or neither at $x = 10$? Justify your answer.
- Does the graph of g have a point of inflection at $x = 4$? Justify your answer.
- Find the absolute minimum value and the absolute maximum value of g on the interval $-4 \leq x \leq 12$. Justify your answers.
- For $-4 \leq x \leq 12$, find all intervals for which $g(x) \leq 0$.

Part A

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for answer with justification

The function g has neither a relative minimum nor a relative maximum at $x = 10$ since $g'(x) = f(x)$ and $f(x) \leq 0$ for $8 \leq x \leq 12$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for answer with justification

The function g has neither a relative minimum nor a relative maximum at $x = 10$ since $g'(x) = f(x)$ and $f(x) \leq 0$ for $8 \leq x \leq 12$.

Part B

5-6b

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for answer with justification

The graph of g has a point of inflection at $x = 4$ since $g'(x) = f(x)$ is increasing for $2 \leq x \leq 4$ and decreasing for $4 \leq x \leq 8$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for answer with justification

The graph of g has a point of inflection at $x = 4$ since $g'(x) = f(x)$ is increasing for $2 \leq x \leq 4$ and decreasing for $4 \leq x \leq 8$.

Part C

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for considering $x = -2$ and $x = 6$ as candidates

$g'(x) = f(x)$ changes sign only at $x = -2$ and $x = 6$

1 point is earned for considering $x = -4$ and $x = 12$

x	$g(x)$
-4	-4
-2	-8
6	8
12	-4

2 points are earned for answers with justification

On the interval $-4 \leq x \leq 12$, the absolute minimum value is $g(-2) = -8$ and the absolute maximum value is $g(6) = 8$.



0	1	2	3	4	5
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5-6b

The student response earns all of the following points:

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for considering $x = -2$ and $x = 6$ as candidates

$g'(x) = f(x)$ changes sign only at $x = -2$ and $x = 6$

1 point is earned for considering $x = -4$ and $x = 12$

x	$g(x)$
-4	-4
-2	-8
6	8
12	-4

2 points are earned for answers with justification

On the interval $-4 \leq x \leq 12$, the absolute minimum value is $g(-2) = -8$ and the absolute maximum value is $g(6) = 8$.

Part D

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

2 point are earned for intervals

$g(x) \leq 0$ for $-4 \leq x \leq 2$ and $10 \leq x \leq 12$.



0	1	2	3
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The student response earns all of the following points:

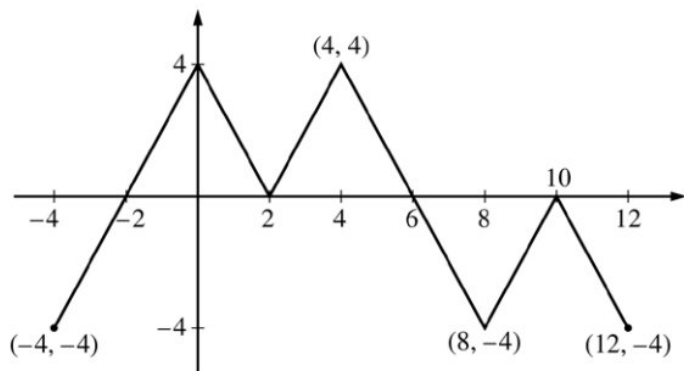
1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

2 point are earned for intervals

$g(x) \leq 0$ for $-4 \leq x \leq 2$ and $10 \leq x \leq 12$.

5-6b

10.

Graph of f

The figure above shows the graph of the piecewise-linear function f . For $-4 \leq x \leq 12$, the function g is defined by $g(x) = \int_{-4}^x f(t) dt$.

Does the graph of g have a point of inflection at $x = 4$? Justify your answer.

Part B

1 point is earned for: answer with justification

The graph of g has a point of inflection at $x = 4$ since $g'(x) = f(x)$ is increasing for $2 \leq x \leq 4$ and decreasing for $4 \leq x \leq 8$.



0

1

Student response earns 1 of the following 1 point(s)

1 point is earned for: answer with justification

The graph of g has a point of inflection at $x = 4$ since $g'(x) = f(x)$ is increasing for $2 \leq x \leq 4$ and decreasing for $4 \leq x \leq 8$.

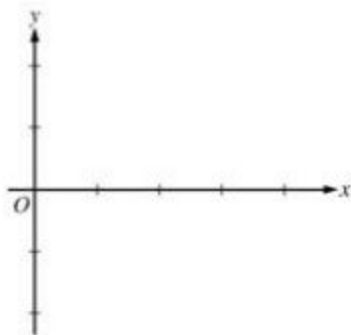
5-6b

11.

x	0	$0 < x < 1$	1	$1 < x < 2$	2	$2 < x < 3$	3	$3 < x < 4$
$f(x)$	-1	Negative	0	Positive	2	Positive	0	Negative
$f'(x)$	4	Positive	0	Positive	DNE	Negative	-3	Negative
$f''(x)$	-2	Negative	0	Positive	DNE	Negative	0	Positive

Let f be a function that is continuous on the interval $[0, 4)$. The function f is twice differentiable except at $x = 2$. The function f and its derivatives have the properties indicated in the table above, where DNE indicates that the derivatives of f do not exist at $x = 2$.

- (a) For $0 < x < 4$, find all values of x at which f has a relative extremum. Determine whether f has a relative maximum or a relative minimum at each of these values. Justify your answer.
- (b) On the axes provided, sketch the graph of a function that has all the characteristics of f .



- (c) Let g be the function defined by $g(x) = \int_1^x f(t) dt$ on the open interval $(0, 4)$. For $0 < x < 4$, find all values of x at which g has a relative extremum. Determine whether g has a relative maximum or a relative minimum at each of these values. Justify your answer.
- (d) For the function g defined in part (c), find all values of x , for $0 < x < 4$, at which the graph of g has a point of inflection. Justify your answer.

Part A

1 point is earned for the relative extremum at $x = 2$

1 point is earned for the relative maximum with justification

f has a relative maximum at $x = 2$ because f' changes from positive to negative at $x = 2$.



0	1	2
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The student response earns all of the following points:

1 point is earned for the relative extremum at $x = 2$

5-6b

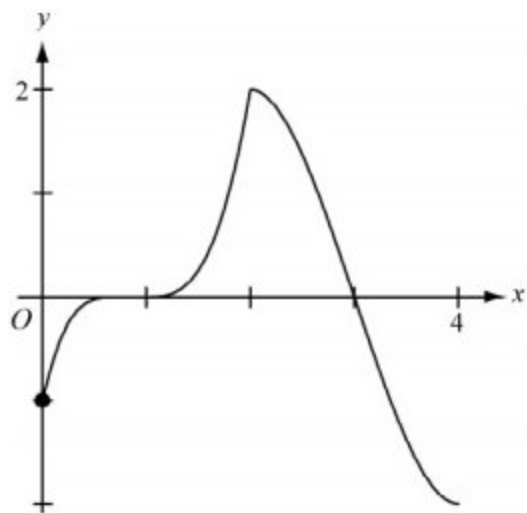
1 point is earned for the relative maximum with justification

f has a relative maximum at $x = 2$ because f' changes from positive to negative at $x = 2$.

Part B

1 point is earned for points at $x = 0, 1, 2, 3$ and behavior at $(2, 2)$

1 point is earned for appropriate increasing/decreasing and concavity behavior



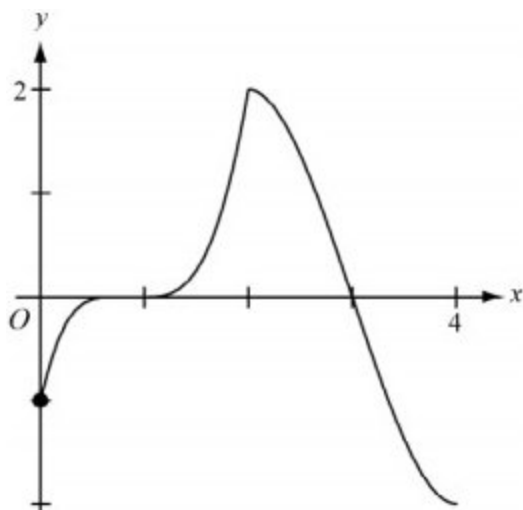
0	1	2
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The student response earns all of the following points:

1 point is earned for points at $x = 0, 1, 2, 3$ and behavior at $(2, 2)$

1 point is earned for appropriate increasing/decreasing and concavity behavior

5-6b

**Part C**

1 point is earned for: $g'(x) = f(x)$

1 point is earned for the critical points

1 point is earned for the answer with justification

$g'(x) = f(x) = 0$ at $x = 1, 3$.

g' changes from negative to positive at $x = 1$ so g has a relative minimum at $x = 1$. g' changes from positive to negative at $x = 3$ so g has a relative maximum at $x = 3$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $g'(x) = f(x)$

1 point is earned for the critical points

1 point is earned for the answer with justification

$g'(x) = f(x) = 0$ at $x = 1, 3$.

g' changes from negative to positive at $x = 1$ so g has a relative minimum at $x = 1$. g' changes from positive to negative at $x = 3$ so g has a relative maximum at $x = 3$.

Part D

5-6b

1 point is earned for: $x = 2$

1 point is earned for the answer with justification

The graph of g has a point of inflection at $x = 2$ because $g'' = f'$ changes sign at $x = 2$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $x = 2$

1 point is earned for the answer with justification

The graph of g has a point of inflection at $x = 2$ because $g'' = f'$ changes sign at $x = 2$.

5-6b

12.

x	0	$0 < x < 1$	1	$1 < x < 2$	2	$2 < x < 3$	3	$3 < x < 4$
$f(x)$	-1	Negative	0	Positive	2	Positive	0	Negative
$f'(x)$	4	Positive	0	Positive	DNE	Negative	-3	Negative
$f''(x)$	-2	Negative	0	Positive	DNE	Negative	0	Positive

Let f be a function that is continuous on the interval $[0, 4)$. The function f is twice differentiable except at $x = 2$. The function f and its derivatives have the properties indicated in the table above, where DNE indicates that the derivatives of f do not exist at $x = 2$.

For the function g defined in part (c), find all values of x , for $0 < x < 4$, at which the graph of g has a point of inflection. Justify your answer

Part D

1 point is earned for: $x = 2$

1 point is earned for the answer with justification

The graph of g has a point of inflection at $x = 2$ because $g'' = f''$

changes sign at $x = 2$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $x = 2$

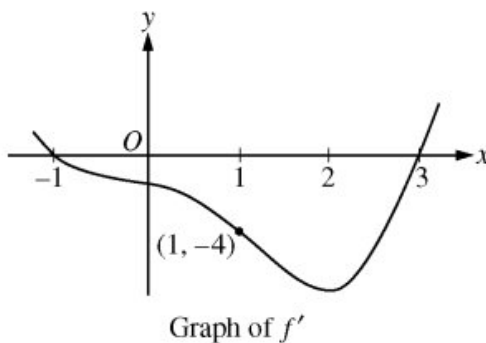
1 point is earned for the answer with justification

The graph of g has a point of inflection at $x = 2$ because $g'' = f''$

changes sign at $x = 2$.

5-6b

13.



Let f be a twice-differentiable function defined on the interval $-1.2 < x < 3.2$ with $f(1) = 2$. The graph of f' , the derivative of f , is shown above. The graph of f' crosses the x -axis at $x = -1$ and $x = 3$ and has a horizontal tangent at $x = 2$. Let g be the function given by $g(x) = e^{f(x)}$.

- Write an equation for the line tangent to the graph of g at $x = 1$.
- For $-1.2 < x < 3.2$, find all values of x at which g has a local maximum. Justify your answer.
- The second derivative of g is $g''(x) = e^{f(x)} [(f'(x))^2 + f''(x)]$. Is $g''(-1)$ positive, negative, or zero? Justify your answer.
- Find the average rate of change of g' , the derivative of g , over the interval $[1, 3]$.

Part A

1 point is earned for $g'(x)$

1 point is earned for $g(1)$ and $g'(1)$

1 point is earned for tangent line equation

$$g(1) = e^{f(1)} = e^2$$

$$g'(x) = e^{f(x)} f'(x), \quad g'(1) = e^{f(1)} f'(1) = -4e^2$$

The tangent line is given by $y = e^2 - 4e^2(x - 1)$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g'(x)$

1 point is earned for $g(1)$ and $g'(1)$

5-6b

1 point is earned for tangent line equation

$$g(1) = e^{f(1)} = e^2$$

$$g'(x) = e^{f(x)} f'(x), \quad g'(1) = e^{f(1)} f'(1) = -4e^2$$

The tangent line is given by $y = e^2 - 4e^2(x - 1)$.

Part B

1 point is earned for the answer

1 point is earned for the justification

$$g'(x) = e^{f(x)} f'(x)$$

$$e^{f(x)} > 0 \text{ for all } x$$

So, g' changes from positive to negative only when f' changes from positive to negative. This occurs at $x = -1$ only. Thus, g has a local maximum at $x = -1$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$$g'(x) = e^{f(x)} f'(x)$$

$$e^{f(x)} > 0 \text{ for all } x$$

So, g' changes from positive to negative only when f' changes from positive to negative. This occurs at $x = -1$ only. Thus, g has a local maximum at $x = -1$.

Part C

5-6b

1 point is earned for the answer

1 point is earned for the justification

$$g''(-1) = e^{f(-1)} \left[(f'(-1))^2 + f''(-1) \right]$$

$$e^{f(-1)} > 0 \text{ and } f'(-1) = 0$$

Since f' is decreasing on a neighborhood of -1 , $f''(-1) < 0$. Therefore, $g''(-1) < 0$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$$g''(-1) = e^{f(-1)} \left[(f'(-1))^2 + f''(-1) \right]$$

$$e^{f(-1)} > 0 \text{ and } f'(-1) = 0$$

Since f' is decreasing on a neighborhood of -1 , $f''(-1) < 0$. Therefore, $g''(-1) < 0$.

Part D

1 point is earned for the difference quotient

1 point is earned for the answer

$$\frac{g'(3) - g'(1)}{3 - 1} = \frac{e^{f(3)} f'(3) - e^{f(1)} f'(1)}{2} = 2e^2$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the difference quotient

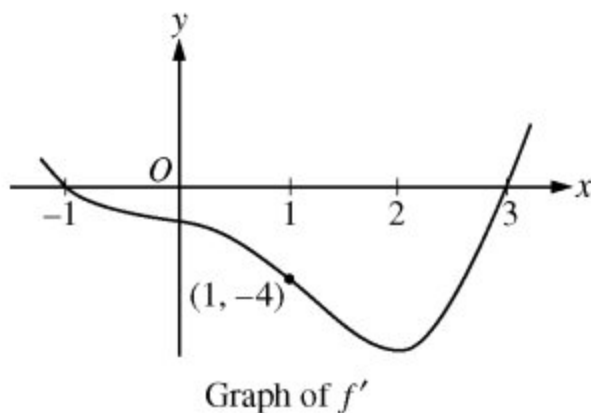
5-6b

1 point is earned for the answer

$$\frac{g'(3) - g'(1)}{3 - 1} = \frac{e^{f(3)} f'(3) - e^{f(1)} f'(1)}{2} = 2e^2$$

5-6b

14.



Let f be a twice-differentiable function defined on the interval $-1.2 < x < 3.2$ with $f(1) = 2$. The graph of f' , the derivative of f , is shown above. The graph of f' crosses the x -axis at $x = -1$ and $x = 3$ and has a horizontal tangent at $x = 2$. Let g be the function given by $g(x) = e^{f(x)}$.

The second derivative of g is $g''(x) = e^{f(x)}[(f'(x))^2 + f''(x)]$. Is $g''(-1)$ positive, negative, or zero? Justify your answer.

Part C

1 point is earned for the answer

1 point is earned for the justification

$$g''(-1) = e^{f(-1)} [(f'(-1))^2 + f''(-1)]$$

$$e^{f(-1)} > 0 \text{ and } f'(-1) = 0$$

Since f' is decreasing on a neighborhood of -1 , $f''(-1) < 0$. Therefore, $g''(-1) < 0$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

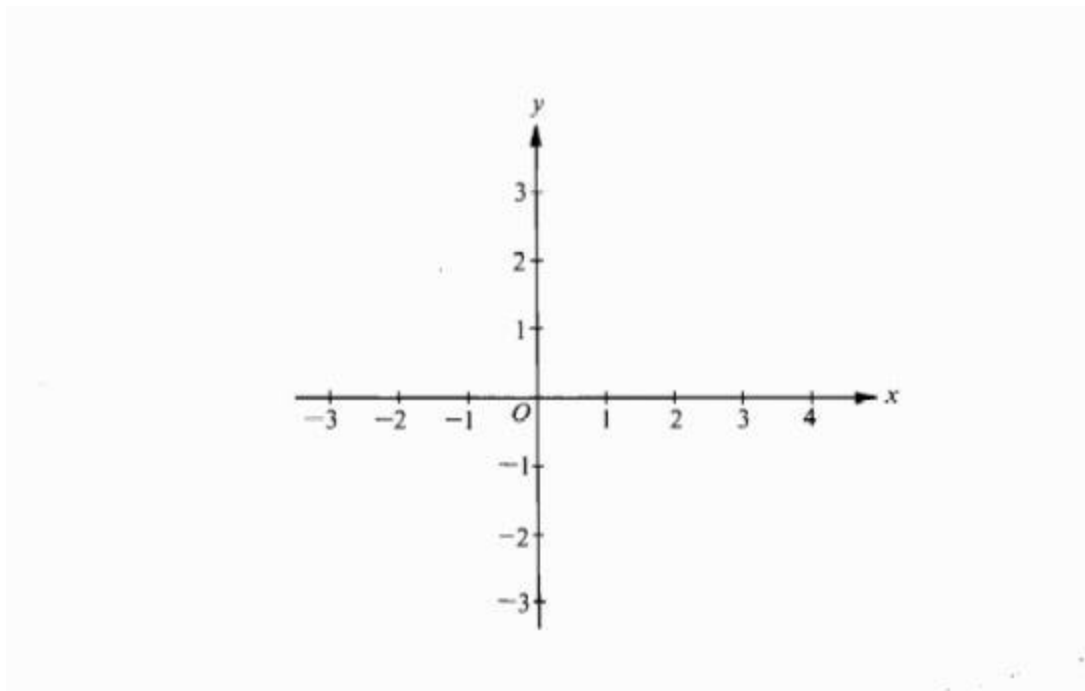
$$g''(-1) = e^{f(-1)} [(f'(-1))^2 + f''(-1)]$$

$$e^{f(-1)} > 0 \text{ and } f'(-1) = 0$$

Since f' is decreasing on a neighborhood of -1 , $f''(-1) < 0$. Therefore, $g''(-1) < 0$.

5-6b

15. Let f be the function defined by $f(x) = 2xe^{-x}$ for all real numbers x .
- (a) Write an equation of the horizontal asymptote for the graph of f .
- (b) Find the x -coordinate of each critical point of f . For each such x , determine whether $f(x)$ is a relative maximum, a relative minimum, or neither.
- (c) For what values of x is the graph of f concave down?
- (d) Using the results found in parts (a), (b), and (c), sketch the graph of $y = f(x)$ in the xy -plane provided below.

**Part A**

1 point is earned for the correct equation

$$y = 0$$



0

1

The student response earns all of the following points:

1 point is earned for the correct equation

$$y = 0$$

Part B

1 point is earned for correct derivative

5-6b

1 point is earned for critical value for f'

1 point is earned for identifying critical point as relative maximum

$$\begin{aligned} f'(x) &= 2(-xe^{-x} + e^{-x}) \\ &= 2e^{-x}(1 - x) \end{aligned}$$

critical point at $x = 1$

relative maximum at $x = 1$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for correct derivative

1 point is earned for critical value for f'

1 point is earned for identifying critical point as relative maximum

$$\begin{aligned} f'(x) &= 2(-xe^{-x} + e^{-x}) \\ &= 2e^{-x}(1 - x) \end{aligned}$$

critical point at $x = 1$

relative maximum at $x = 1$

Part C

1 point is earned for correct $f''(x)$ for $f'(x)$ found in (b)

1 point is earned for correct interval

$$\begin{aligned} f''(x) &= 2e^{-x}(-1) + (-2e^{-x})(1 - x) \\ &= 2e^{-x}(x - 2) \end{aligned}$$

Concave down when

5-6b

$$2e^{-x}(x-2) < 0$$

$$(x-2) < 0$$

$$x < 2$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correct $f''(x)$ for $f'(x)$ found in (b)

1 point is earned for correct interval

$$\begin{aligned} f''(x) &= 2e^{-x}(-1) + (-2e^{-x})(1-x) \\ &= 2e^{-x}(x-2) \end{aligned}$$

Concave down when

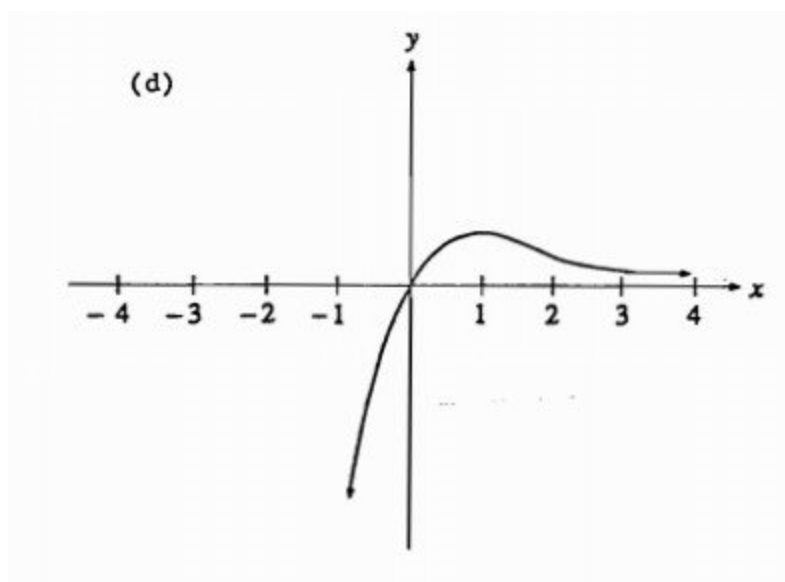
$$2e^{-x}(x-2) < 0$$

$$(x-2) < 0$$

$$x < 2$$

Part D

3 points are earned for the graph consistent with information found in (a), (b), and (c)



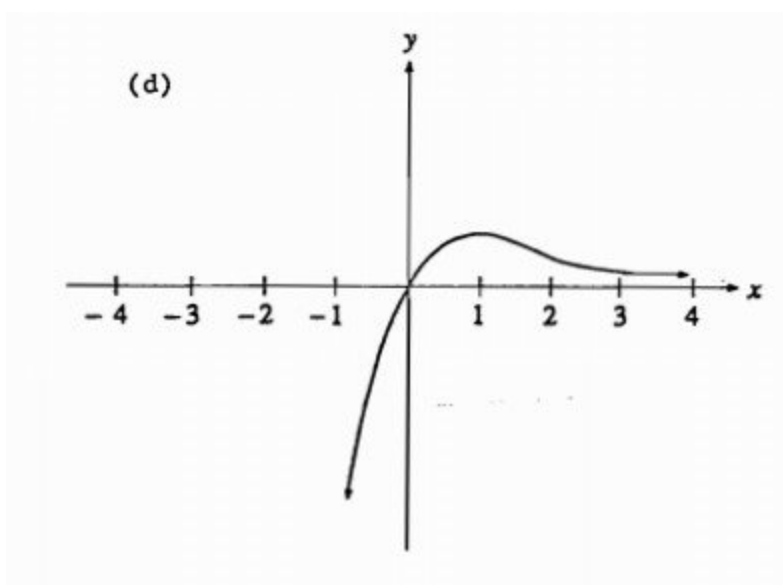
5-6b



0	1	2	3
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The student response earns all of the following points:

3 points are earned for the graph consistent with information found in (a), (b), and (c)



5-6b

16. Let F be the function defined by $f(x) = 2xe^{-x}$ for all real numbers x .
For what values of x is the graph of f concave down?

Part C

1 point is earned for correct $f''(x)$ for $f'(x)$ found in (b)

1 point is earned for correct interval

$$\begin{aligned} f''(x) &= 2e^{-x}(-1) + (-2e^{-x})(1-x) \\ &= 2e^{-x}(x-2) \end{aligned}$$

Concave down when

$$2e^{-x}(x-2) < 0$$

$$(x-2) < 0$$

$$x < 2$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correct $f''(x)$ for $f'(x)$ found in (b)

1 point is earned for correct interval

$$\begin{aligned} f''(x) &= 2e^{-x}(-1) + (-2e^{-x})(1-x) \\ &= 2e^{-x}(x-2) \end{aligned}$$

Concave down when

$$2e^{-x}(x-2) < 0$$

$$(x-2) < 0$$

$$x < 2$$

5-6b

17. Let f be a twice-differentiable function such that $f(2) = 5$ and $f(5) = 2$. Let g be the function given by $g(x) = f(f(x))$.
- (a) Explain why there must be a value c for $2 < c < 5$ such that $f'(c) = -1$.
- (b) Show that $g'(2) = g'(5)$. Use this result to explain why there must be a value k for $2 < k < 5$ such that $g''(k) = 0$.
- (c) Show that if $f''(x) = 0$ for all x , then the graph of g does not have a point of inflection.
- (d) Let $h(x) = f(x) - x$. Explain why there must be a value r for $2 < r < 5$ such that $h(r) = 0$.

Part A

1 point is earned for $\frac{f(5) - f(2)}{5 - 2}$

1 point is earned for conclusion, using MVT

The Mean Value Theorem guarantees that there is a value c , with $2 < c < 5$, so that

$$f'(c) = \frac{f(5) - f(2)}{5 - 2} = \frac{2 - 5}{5 - 2} = -1.$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{f(5) - f(2)}{5 - 2}$

1 point is earned for conclusion, using MVT

The Mean Value Theorem guarantees that there is a value c , with $2 < c < 5$, so that

$$f'(c) = \frac{f(5) - f(2)}{5 - 2} = \frac{2 - 5}{5 - 2} = -1.$$

Part B

1 point is earned for $g'(x)$

5-6b

1 point is earned for $g'(2) = f'(5) \cdot f'(2) = g'(5)$

1 point is earned if uses MVT with g'

$$g'(x) = f'(f(x)) \cdot f'(x)$$

$$g'(2) = f'(f(2)) \cdot f'(2) = f'(5) \cdot f'(2)$$

$$g'(5) = f'(f(5)) \cdot f'(5) = f'(2) \cdot f'(5)$$

Thus, $g'(2) = g'(5)$.

Since f is twice-differentiable, g' is differentiable everywhere, so the Mean Value Theorem applied to g' on $[2, 5]$

guarantees there is a value k , with $2 < k < 5$, such that $g''(k) = \frac{g'(5) - g'(2)}{5 - 2} = 0$.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for $g'(x)$

1 point is earned for $g'(2) = f'(5) \cdot f'(2) = g'(5)$

1 point is earned if uses MVT with g'

$$g'(x) = f'(f(x)) \cdot f'(x)$$

$$g'(2) = f'(f(2)) \cdot f'(2) = f'(5) \cdot f'(2)$$

$$g'(5) = f'(f(5)) \cdot f'(5) = f'(2) \cdot f'(5)$$

Thus, $g'(2) = g'(5)$.

Since f is twice-differentiable, g' is differentiable everywhere, so the Mean Value Theorem applied to g' on $[2, 5]$

guarantees there is a value k , with $2 < k < 5$, such that $g''(k) = \frac{g'(5) - g'(2)}{5 - 2} = 0$.

Part C

1 point is earned if considers g''

1 point is earned for $g''(x) = 0$ for all x

5-6b

OR

1 point is earned for f is linear1 point is earned for g is linear

$$g''(x) = f''(f(x)) \cdot f'(x) \cdot f'(x) + f'(f(x)) \cdot f''(x)$$

If $f''(x) = 0$ for all x , then

$$g''(x) = 0 \cdot f'(x) \cdot f'(x) + f'(f(x)) \cdot 0 = 0 \text{ for all } x.$$

Thus, there is no x -value at which $g''(x)$ changes sign, so the graph of g has no inflection points. ORIf $f''(x) = 0$ for all x , then f is linear, so $g = f \circ f$ is linear and the graph of g has no inflection points.

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned if considers g'' 1 point is earned for $g''(x) = 0$ for all x

OR

1 point is earned for f is linear1 point is earned for g is linear

$$g''(x) = f''(f(x)) \cdot f'(x) \cdot f'(x) + f'(f(x)) \cdot f''(x)$$

If $f''(x) = 0$ for all x , then

$$g''(x) = 0 \cdot f'(x) \cdot f'(x) + f'(f(x)) \cdot 0 = 0 \text{ for all } x.$$

Thus, there is no x -value at which $g''(x)$ changes sign, so the graph of g has no inflection points. ORIf $f''(x) = 0$ for all x , then f is linear, so $g = f \circ f$ is linear and the graph of g has no inflection points.**Part D**1 point is earned for $h(2)$ and $h(5)$

1 point is earned for conclusion, using IVT

Let $h(x) = f(x) - x$.

5-6b

$$h(2) = f(2) - 2 = 5 - 2 = 3$$

$$h(5) = f(5) - 5 = 2 - 5 = -3$$

Since $h(2) > 0 > h(5)$, the Intermediate Value Theorem guarantees that there is a value r , with $2 < r < 5$, such that $h(r) = 0$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $h(2)$ and $h(5)$

1 point is earned for conclusion, using IVT

Let $h(x) = f(x) - x$.

$$h(2) = f(2) - 2 = 5 - 2 = 3$$

$$h(5) = f(5) - 5 = 2 - 5 = -3$$

Since $h(2) > 0 > h(5)$, the Intermediate Value Theorem guarantees that there is a value r , with $2 < r < 5$, such that $h(r) = 0$.

5-6b

18. Let f be a twice-differentiable function such that $f(2)=5$ and $f(5)=2$. Let g be the function given by $g(x)=f(f(x))$.

Show that if $f''(x)=0$ for all x , then the graph of g does not have a point of inflection.

Part C

1 point is earned if considers g''

1 point is earned for $g''(x)=0$ for all x

OR

1 point is earned for f is linear

1 point is earned for g is linear

$$g''(x) = f''(f(x)) * f'(x) * f'(x) + f'(f(x)) * f''(x)$$

If $f''(x)=0$ for all x , then

$$g''(x)=0 * f'(x) * f'(x) + f'(f(x)) * 0 = 0 \text{ for all } x.$$

Thus, there is no x -value at which $g''(x)$ changes sign, so the graph of g has no inflection points.

OR

If $f''(x)=0$ for all x , then f is linear, so $g = f \circ f$ is linear and the graph of g has no inflection points.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned if considers g''

1 point is earned for $g''(x)=0$ for all x

OR

1 point is earned for f is linear

1 point is earned for g is linear

$$g''(x) = f''(f(x)) * f'(x) * f'(x) + f'(f(x)) * f''(x)$$

If $f''(x)=0$ for all x , then

5-6b

$$g''(x) = 0 \cdot f'(x) \cdot f'(x) + f'(f(x)) \cdot 0 = 0 \text{ for all } x.$$

Thus, there is no x -value at which $g''(x)$ changes sign, so the graph of g has no inflection points.

OR

If $f'(x) = 0$ for all x , then f is linear, so $g = f \circ f$ is linear and the graph of g has no inflection points.

5-6b

19. A cubic polynomial function f is defined by

$$f(x) = 4x^3 + ax^2 + bx + k$$

where a , b , and k are constants. The function f has a local minimum at $x = -1$, and the graph f has a point of inflection at $x = -2$.

(a) Find the values of a and b .

(b) If $\int_0^1 f(x) dx = 32$, what is the value of k ?

Part A

1 point is earned for: $f'(x)$

1 point is earned for: $f''(x)$

1 point is earned for: $f'(-1) = 0$

1 point is earned for: $f''(-2) = 0$

1 point is earned for: a , b

$$f'(x) = 12x^2 + 2ax + b$$

$$f''(x) = 24x + 2a$$

$$f'(-1) = 12 - 2a + b = 0$$

$$f''(-2) = -48 + 2a = 0$$

$$a = 24$$

$$b = -12 + 2a = 36$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for: $f'(x)$

1 point is earned for: $f''(x)$

1 point is earned for: $f'(-1) = 0$

1 point is earned for: $f''(-2) = 0$

5-6b

1 point is earned for: a, b

$$f'(x) = 12x^2 + 2ax + b$$

$$f''(x) = 24x + 2a$$

$$f'(-1) = 12 - 2a + b = 0$$

$$f''(-2) = -48 + 2a = 0$$

$$a = 24$$

$$b = -12 + 2a = 36$$

Part B

2 points are earned for antidifferentiation <-1> each error

1 point is earned for the expression in k

1 point is earned for k

$$\int_0^1 (4x^3 + 24x^2 + 36x + k) dx$$

$$= x^4 + 8x^3 + 18x^2 + kx \Big|_{x=0}^{x=1} = 27 + k$$

$$27 + k = 32$$

$$k = 5$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for antidifferentiation <-1> each error

1 point is earned for the expression in k

1 point is earned for k

5-6b

$$\begin{aligned}\int_0^1 (4x^3 + 24x^2 + 36x + k) dx \\&= x^4 + 8x^3 + 18x^2 + kx \Big|_{x=0}^{x=1} = 27 + k \\27 + k &= 32 \\k &= 5\end{aligned}$$

5-6b

20. A cubic polynomial function f is defined by

$$f(x) = 4x^3 + ax^2 + bx + k$$

where a , b , and k are constants. The function f has a local minimum at $x = -1$, and the graph f has a point of inflection at $x = -2$.

Find the values of a and b .

Part A

1 point is earned for: $f'(x)$

1 point is earned for: $f''(x)$

1 point is earned for: $f'(-1) = 0$

1 point is earned for: $f''(-2) = 0$

1 point is earned for: a , b

$$f'(x) = 12x^2 + 2ax + b$$

$$f''(x) = 24x + 2a$$

$$f'(-1) = 12 - 2a + b = 0$$

$$f''(-2) = -48 + 2a = 0$$

$$a = 24$$

$$b = -12 + 2a = 36$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for: $f'(x)$

1 point is earned for: $f''(x)$

1 point is earned for: $f'(-1) = 0$

1 point is earned for: $f''(-2) = 0$

1 point is earned for: a , b

$$f'(x) = 12x^2 + 2ax + b$$

5-6b

$$f'(x) = 24x + 2a$$

$$f'(-1) = 12 - 2a + b = 0$$

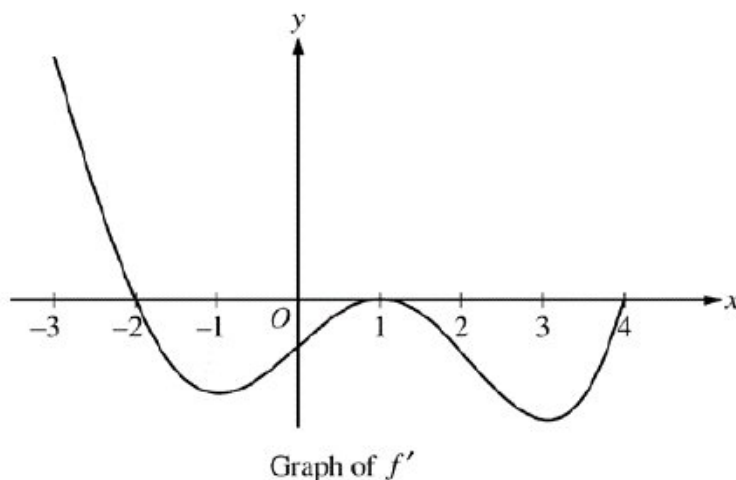
$$f'(-2) = -48 + 2a = 0$$

$$a = 24$$

$$b = -12 + 2a = 36$$

5-6b

21.



The figure above shows the graph of f' , the derivative of a twice-differentiable function f , on the interval $[-3, 4]$. The graph of f' has horizontal tangents at $x = -1$, $x = 1$, and $x = 3$. The areas of the regions bounded by the x -axis and the graph of f' on the intervals $[-2, 1]$ and $[1, 4]$ are 9 and 12, respectively.

- Find all x -coordinates at which f has a relative maximum. Give a reason for your answer.
- On what open intervals contained in $-3 < x < 4$ is the graph of f both concave down and decreasing? Give a reason for your answer.
- Find the x -coordinates of all points of inflection for the graph of f . Give a reason for your answer.
- Given that $f(1) = 3$, write an expression for $f(x)$ that involves an integral. Find $f(4)$ and $f(-2)$.

Part A

1 point is earned for identifying $x = -2$

1 point is earned for answer with reason

$f'(x) = 0$ at $x = -2$, $x = 1$, and $x = 4$.

$f'(x)$ changes from positive to negative at $x = -2$.

Therefore, f has a relative maximum at $x = -2$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for identifying $x = -2$

1 point is earned for answer with reason

$f'(x) = 0$ at $x = -2$, $x = 1$, and $x = 4$.

5-6b

$f(x)$ changes from positive to negative at $x = -2$.

Therefore, f has a relative maximum at $x = -2$.

Part B

1 point is earned for the intervals

1 point is earned for the reason

The graph of f is concave down and decreasing on the intervals $-2 < x < -1$ and $1 < x < 3$ because f' is decreasing and negative on these intervals.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the intervals

1 point is earned for the reason

The graph of f is concave down and decreasing on the intervals $-2 < x < -1$ and $1 < x < 3$ because f' is decreasing and negative on these intervals.

Part C

1 point is earned for identifying $x = -1, 1$, and 3

1 point is earned for the reason

The graph of f has a point of inflection at $x = -1$ and $x = 3$ because f' changes from decreasing to increasing at these points. The graph of f has a point of inflection at $x = 1$ because f' changes from increasing to decreasing at this point.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for identifying $x = -1, 1$, and 3

1 point is earned for the reason

5-6b

The graph of f has a point of inflection at $x = -1$ and $x = 3$ because f' changes from decreasing to increasing at these points. The graph of f has a point of inflection at $x = 1$ because f' changes from increasing to decreasing at this point.

Part D

1 point is earned for the integrand

1 point is earned for expressing $f(x)$

1 point is earned for $f(4)$ and $f(-2)$

$$f(x) = 3 + \int_1^x f'(t) dt$$

$$f(4) = 3 + \int_1^4 f'(t) dt = 3 + (-12) = -9$$

$$\begin{aligned} f(-2) &= 3 + \int_1^{-2} f'(t) dt = 3 - \int_{-2}^1 f'(t) dt \\ &= 3 - (-9) = 12 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for expressing $f(x)$

1 point is earned for $f(4)$ and $f(-2)$

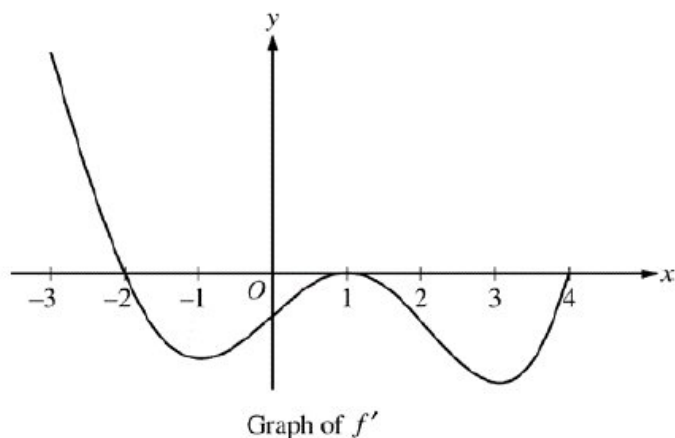
$$f(x) = 3 + \int_1^x f'(t) dt$$

$$f(4) = 3 + \int_1^4 f'(t) dt = 3 + (-12) = -9$$

$$\begin{aligned} f(-2) &= 3 + \int_1^{-2} f'(t) dt = 3 - \int_{-2}^1 f'(t) dt \\ &= 3 - (-9) = 12 \end{aligned}$$

5-6b

22.



The figure above shows the graph of f' , the derivative of a twice-differentiable function f , on the interval $[-3, 4]$. The graph of f' has horizontal tangents at $x = -1$, $x = 1$, and $x = 3$. The areas of the regions bounded by the x -axis and the graph of f' on the intervals $[-2, 1]$ and $[1, 4]$ are 9 and 12, respectively.

On what open intervals contained in $-3 < x < 4$ is the graph of f both concave down and decreasing? Give a reason for your answer.

Part B

1 point is earned for the intervals

1 point is earned for the explanation

The graph of f is concave down and decreasing on the intervals $-2 < x < -1$ and $1 < x < 3$ because f' is decreasing and negative on these intervals.



0	1	2
---	---	---

The student response earns all of the following points:

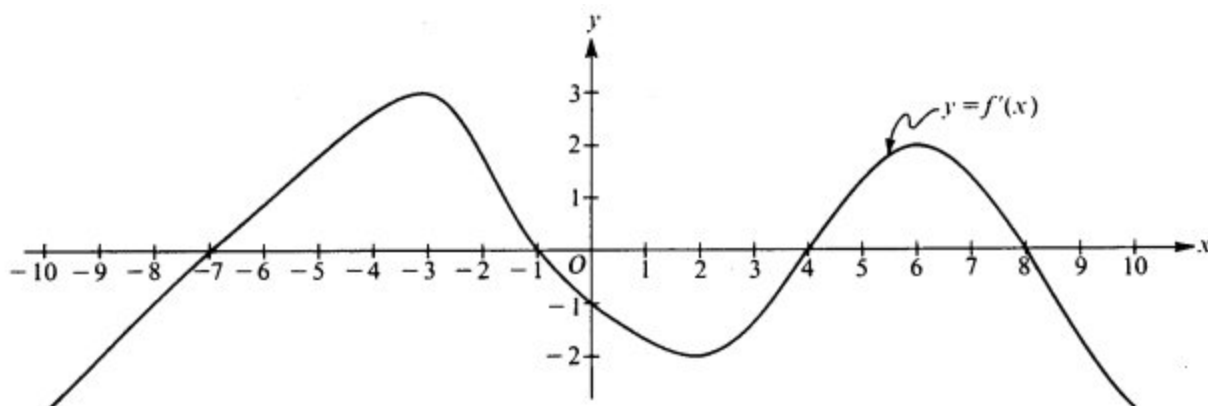
1 point is earned for the intervals

1 point is earned for the explanation

The graph of f is concave down and decreasing on the intervals $-2 < x < -1$ and $1 < x < 3$ because f' is decreasing and negative on these intervals.

5-6b

23.



Note: This is the graph of the derivative of f , not the graph of f .

The figure above shows the graph of f' , the derivative of a function f . The domain of f is the set of all real numbers x such that $-10 \leq x \leq 10$.

- For what values of x does the graph of f have a horizontal tangent?
- For what values of x in the interval $(-10, 10)$ does f have a relative maximum? Justify your answer
- For what values of x is the graph of f concave downwards?

Part A

3 point is earned for solving $f'(x) = 0$

1 point is **deducted** for each incorrect point

1 point is **deducted** for each missing point

horizontal tangent $\Leftrightarrow f'(x) = 0$

$x = -7, -1, 4, 8$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

3 point is earned for solving $f'(x) = 0$

1 point is **deducted** for each incorrect point

1 point is **deducted** for each missing point

horizontal tangent $\Leftrightarrow f'(x) = 0$

$x = -7, -1, 4, 8$

5-6b

Part B

2 points are earned for answer

1 point is earned for justification

1 point is **deducted** for each incorrect or missing point

Relative maxima at $x = -1, 8$ because f' changes from positive to negative at these points



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for answer

1 point is earned for justification

1 point is **deducted** for each incorrect or missing point

Relative maxima at $x = -1, 8$ because f' changes from positive to negative at these points

Part C

1 of 3 points are earned for correct sub intervals

2 of 3 points are earned for 1 correct interval with analysis

3 of 3 points are earned for 2 correct intervals

f concave downward $\Rightarrow f'$ decreasing $(-3, 2), (6, 10)$



0	1	2	3
---	---	---	---

The student response earns all points:

1 of 3 points are earned for correct sub intervals

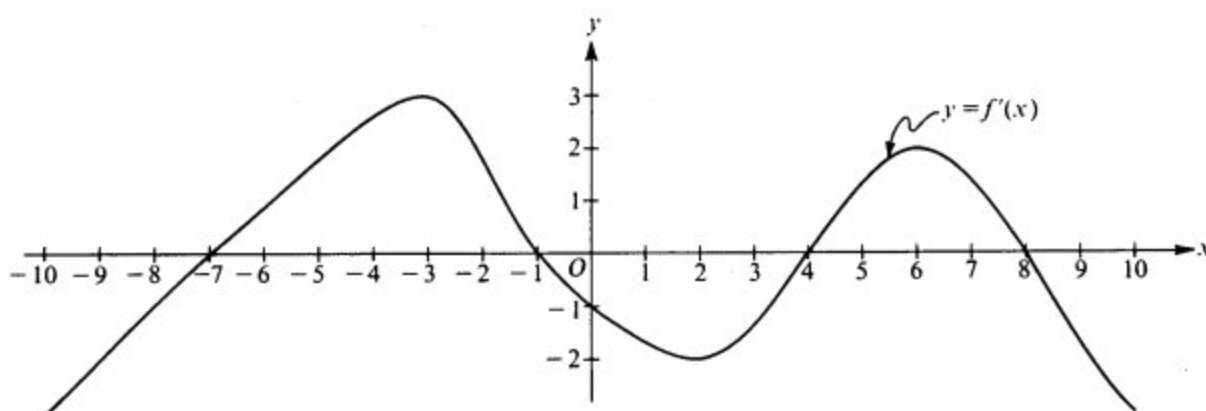
2 of 3 points are earned for 1 correct interval with analysis

3 of 3 points are earned for 2 correct intervals

f concave downward $\Rightarrow f'$ decreasing $(-3, 2), (6, 10)$

5-6b

24.



Note: This is the graph of the derivative of f , not the graph of f .

The figure above shows the graph of f' , the derivative of a function f . The domain of f is the set of all real numbers x such that $-10 \leq x \leq 10$.

For what values of x is the graph of f concave downwards?

Part C

1 of 3 points are earned for correct sub intervals

2 of 3 points are earned for 1 correct interval with analysis

3 of 3 points are earned for 2 correct intervals

f concave downward $\Leftrightarrow f'$ decreasing

$(-3, 2), (6, 10)$



0	1	2	3
---	---	---	---

The student response earns three points:

1 of 3 points are earned for correct sub intervals

2 of 3 points are earned for 1 correct interval with analysis

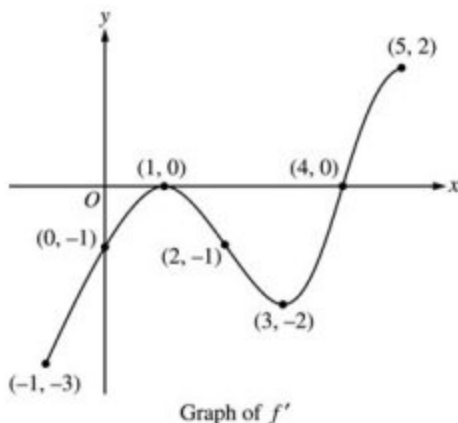
3 of 3 points are earned for 2 correct intervals

f concave downward $\Leftrightarrow f'$ decreasing

$(-3, 2), (6, 10)$

5-6b

25.



The figure above shows the graph of f' , the derivative of the function f , on the closed interval $-1 \leq x \leq 5$. The graph of f' has horizontal tangent lines at $x = 1$ and $x = 3$. The function f is twice differentiable with $f(2) = 6$.

- (a) Find the x -coordinate of each of the points of inflection of the graph of f . Give a reason for your answer.
- (b) At what value of x does f attain its absolute minimum value on the closed interval $-1 \leq x \leq 5$? At what value of x does f attain its absolute maximum value on the closed interval $-1 \leq x \leq 5$? Show the analysis that leads to your answers.
- (c) Let g be the function defined by $g(x) = xf(x)$. Find an equation for the line tangent to the graph of g at $x = 2$.

Part A

1 point is earned for $x = 1, x = 3$

1 point is earned for the reason

$x = 1$ and $x = 3$ because the graph of f' changes from increasing to decreasing at $x = 1$, and changes from decreasing to increasing at $x = 3$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $x = 1, x = 3$

1 point is earned for the reason

$x = 1$ and $x = 3$ because the graph of f' changes from increasing to decreasing at $x = 1$, and changes from decreasing to increasing at $x = 3$.

Part B

5-6b

1 point is earned for indicating f decreases then increases

1 point is earned for eliminating $x = 5$ for maximum

1 point is earned for absolute minimum at $x = 4$

1 point is earned for absolute maximum at $x = -1$

The function f decreases from $x = -1$ to $x = 4$, then increases from $x = 4$ to $x = 5$.

Therefore, the absolute minimum value for f is at $x = 4$. The absolute maximum value must occur at $x = -1$ or at $x = 5$.

$$f(5) - f(-1) = \int_{-1}^5 f'(t) dt < 0$$

Since $f(5) < f(-1)$, the absolute maximum value occurs at $x = -1$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for indicating f decreases then increases

1 point is earned for eliminating $x = 5$ for maximum

1 point is earned for absolute minimum at $x = 4$

1 point is earned for absolute maximum at $x = -1$

The function f decreases from $x = -1$ to $x = 4$, then increases from $x = 4$ to $x = 5$.

Therefore, the absolute minimum value for f is at $x = 4$. The absolute maximum value must occur at $x = -1$ or at $x = 5$.

$$f(5) - f(-1) = \int_{-1}^5 f'(t) dt < 0$$

Since $f(5) < f(-1)$, the absolute maximum value occurs at $x = -1$.

Part C

2 points are earned for $g'(x)$

1 point is earned for the tangent line

5-6b

$$g'(x) = f(x) + xf'(x)$$

$$g'(2) = f(2) + 2f'(2) = 6 + 2(-1) = 4$$

$$g(2) = 2f(2) = 12$$

$$\text{Tangent line is } y = 4(x - 2) + 12$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $g'(x)$

1 point is earned for the tangent line

$$g'(x) = f(x) + xf'(x)$$

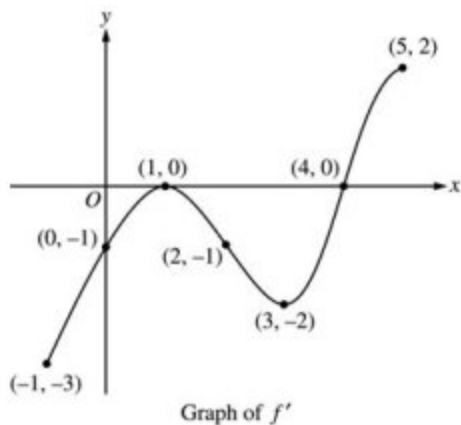
$$g'(2) = f(2) + 2f'(2) = 6 + 2(-1) = 4$$

$$g(2) = 2f(2) = 12$$

$$\text{Tangent line is } y = 4(x - 2) + 12$$

5-6b

26.



The figure above shows the graph of f' , the derivative of the function f , on the closed interval $-1 \leq x \leq 5$. The graph of f' has horizontal tangent lines at $x=1$ and $x=3$. The function f is twice differentiable with $f(2)=6$.

Find the x -coordinate of each of the points of inflection of the graph of f . Give a reason for your answer.

Part A

1 point is earned for $x=1, x=3$

1 point is earned for the reason

$x=1$ and $x=3$ because the graph of f' changes from increasing to decreasing at $x=1$, and changes from decreasing to increasing at $x=3$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $x=1, x=3$

1 point is earned for the reason

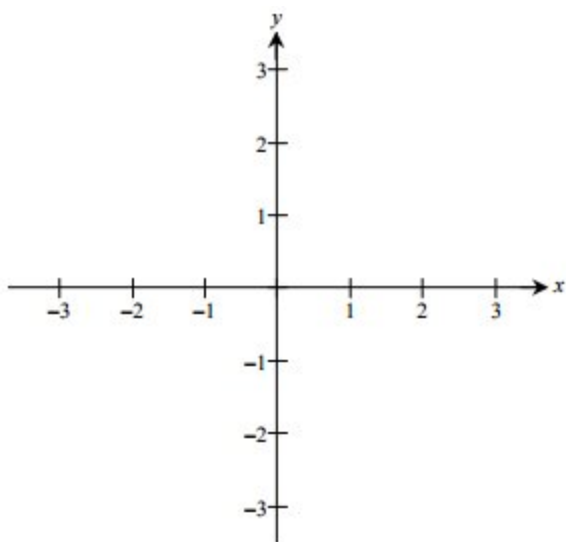
$x=1$ and $x=3$ because the graph of f' changes from increasing to decreasing at $x=1$, and changes from decreasing to increasing at $x=3$.

5-6b

27. Let f be a function that is even and continuous on the closed interval $[-3, 3]$. The function f and its derivatives have the properties indicated in the table below.

x	0	$0 < x < 1$	1	$1 < x < 2$	2	$2 < x < 3$
$f(x)$	1	Positive	0	Negative	-1	Negative
$f'(x)$	Undefined	Negative	0	Negative	Undefined	Positive
$f''(x)$	Undefined	Positive	0	Negative	Undefined	Negative

- (a) Find the x -coordinate of each point at which f attains an absolute maximum value or an absolute minimum value. For each x -coordinate you give, state whether f attains an absolute maximum or an absolute minimum.
- (b) Find the x -coordinate of each point of inflection on the graph of f . Justify your answer.
- (c) In the xy -plane provided below, sketch the graph of a function with all the given characteristics of f .



Part A

1 point is earned for first point

1 point is earned for second point

1 point is earned for third point

1 point is **deducted** for each additional point or incorrect point

Absolute maximum at $x = 0$

Absolute minimum at $x = \pm 2$



0	1	2	3
---	---	---	---

5-6b

The student response earns all of the following points:

1 point is earned for first point

1 point is earned for second point

1 point is earned for third point

1 point is **deducted** for each additional point or incorrect point

Absolute maximum at $x = 0$

Absolute minimum at $x = \pm 2$

Part B

1 point is earned for first point identified

1 point is earned for second point identified

1 point is earned for justification (required on $[0,3)$ only)

1 point is **deducted** for each additional point or incorrect point

The points of inflection are at $x = \pm 1$ because the sign of $f''(x)$ changes at $x = 1$ and f is even.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for first point identified

1 point is earned for second point identified

1 point is earned for justification (required on $[0,3)$ only)

1 point is **deducted** for each additional point or incorrect point

The points of inflection are at $x = \pm 1$ because the sign of $f''(x)$ changes at $x = 1$ and f is even.

Part C

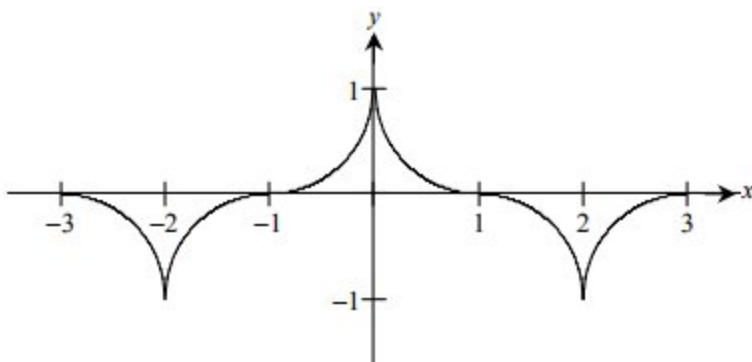
3 points are earned for sketch without error

1 point is **deducted** for each error:

5-6b

- symmetry
- slopes on $[0,3)$
- concavity on $[0,3)$
- discontinuity on $[0,3)$
- $f(0), f(1), f(2)$ incorrect or extra intercept on $[0,3)$

Note: No more than 1 point can be lost in any one category.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

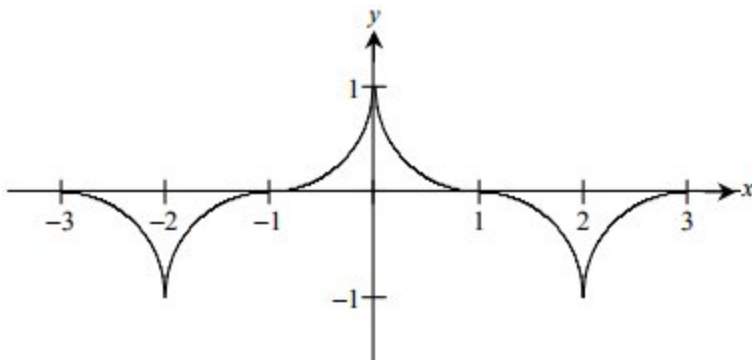
3 points are earned for sketch without error

1 point is **deducted** for each error:

- symmetry
- slopes on $[0,3)$
- concavity on $[0,3)$
- discontinuity on $[0,3)$
- $f(0), f(1), f(2)$ incorrect or extra intercept on $[0,3)$

Note: No more than 1 point can be lost in any one category.

5-6b



28. Let f be a function that is even and continuous on the closed interval $[-3, 3]$. The function f and its derivatives have the properties indicated in the table below.

x	0	$0 < x < 1$	1	$1 < x < 2$	2	$2 < x < 3$
$f(x)$	1	Positive	0	Negative	-1	Negative
$f'(x)$	Undefined	Negative	0	Negative	Undefined	Positive
$f''(x)$	Undefined	Positive	0	Negative	Undefined	Negative

Find the x -coordinate of each point of inflection on the graph of f . Justify your answer.

Part B

1 point is earned for first point identified

1 point is earned for second point identified

1 point is earned for justification (required on $[0, 3)$ only)

1 point is **deducted** for each additional point or incorrect point

The points of inflection are at $x = \pm 1$ because the sign of $f''(x)$ changes at $x = 1$ and f is even.



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for first point identified

1 point is earned for second point identified

1 point is earned for justification (required on $[0, 3)$ only)

1 point is **deducted** for each additional point or incorrect point

The points of inflection are at $x = \pm 1$ because the sign of $f''(x)$ changes at $x = 1$ and f is even.

5-6b

29. Let f be the function given by $f(x) = \frac{\ln x}{x}$ **mfrac** for all $x > 0$. The derivative of f is given by $f'(x) = \frac{1 - \ln x}{x^2}$.

- (a) Write an equation for the line tangent to the graph of f at $x = e^2$.
- (b) Find the x -coordinate of the critical point of f . Determine whether this point is a relative minimum, a relative maximum, or neither for the function f . Justify your answer.
- (c) The graph of the function f has exactly one point of inflection. Find the x -coordinate of this point.
- (d) Find $\lim_{x \rightarrow 0^+} f(x)$.

Part A

1 point is earned for the $f(e^2)$ and $f'(e^2)$

1 point is earned for the answer

$$f(e^2) = \frac{\ln e^2}{e^2} = \frac{2}{e^2}, \quad f'(e^2) = \frac{1 - \ln e^2}{(e^2)^2} = -\frac{1}{e^4}$$

An equation for the tangent line is $y = \frac{2}{e^2} - \frac{1}{e^4}(x - e^2)$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the $f(e^2)$ and $f'(e^2)$

1 point is earned for the answer

$$f(e^2) = \frac{\ln e^2}{e^2} = \frac{2}{e^2}, \quad f'(e^2) = \frac{1 - \ln e^2}{(e^2)^2} = -\frac{1}{e^4}$$

An equation for the tangent line is $y = \frac{2}{e^2} - \frac{1}{e^4}(x - e^2)$.

5-6b

Part B

3 points can be earned.

1 point for $x = e$

1 point for relative maximum

1 point for justification

$f'(x) = 0$ when $x = e$. The function f has a relative maximum at $x = e$ because $f'(x)$ changes from positive to negative at $x = e$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

3 points can be earned.

1 point for $x = e$

1 point for relative maximum

1 point for justification

$f'(x) = 0$ when $x = e$. The function f has a relative maximum at $x = e$ because $f'(x)$ changes from positive to negative at $x = e$.

Part C

2 points are earned for $f''(x)$

1 point is earned for answer

$$f''(x) = \frac{-\frac{1}{x}x^2 - (1 - \ln x)2x}{x^4} = \frac{-3 + 2 \ln x}{x^3} \text{ for all } x > 0$$

$$f''(x) = 0 \text{ when } -3 + 2 \ln x = 0$$

$$x = e^{3/2}$$

The graph of f has a point of inflection at $x = e^{3/2}$ because $f''(x)$ changes sign at $x = e^{3/2}$.

5-6b



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $f''(x)$

1 point is earned for answer

$$f''(x) = \frac{-\frac{1}{x}x^2 - (1 - \ln x)2x}{x^4} = \frac{-3 + 2 \ln x}{x^3} \text{ for all } x > 0$$

$$f''(x) = 0 \text{ when } -3 + 2 \ln x = 0$$

$$x = e^{3/2}$$

The graph of f has a point of inflection at $x = e^{3/2}$ because $f''(x)$ changes sign at $x = e^{3/2}$.

Part D

1 point is earned for answer

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{x} = -\infty \text{ or Does Not Exist}$$



0	1
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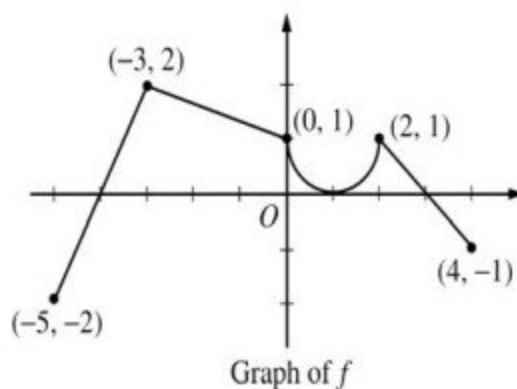
The student response earns all of the following points:

1 point is earned for answer

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{x} = -\infty \text{ or Does Not Exist}$$

5-6b

30.



The graph of the function f shown above consists of a semicircle and three line segments. Let g be the function given by $g(x) = \int_{-3}^x f(t) dt$.

- Find $g(0)$ and $g'(0)$.
- Find all values of x in the open interval $(-5, 4)$ at which g attains a relative maximum. Justify your answer.
- Find the absolute minimum value of g on the closed interval $[-5, 4]$. Justify your answer.
- Find all values of x in the open interval $(-5, 4)$ at which the graph of g has a point of inflection.

Part A

1 point is earned for $g(0)$

1 point is earned for $g'(0)$

$$g(0) = \int_{-3}^0 f(t) dt = \frac{1}{2}(3)(2 + 1) = \frac{9}{2}$$

$$g'(0) = f(0) = 1$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g(0)$

1 point is earned for $g'(0)$

5-6b

$$g(0) = \int_{-3}^0 f(t) dt = \frac{1}{2}(3)(2+1) = \frac{9}{2}$$

$$g'(0) = f(0) = 1$$

Part B

1 point is earned for $x = 3$

1 point is earned for justification

g has a relative maximum at $x = 3$.

This is the only x -value where $g' = f$ changes from positive to negative.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $x = 3$

1 point is earned for justification

g has a relative maximum at $x = 3$.

This is the only x -value where $g' = f$ changes from positive to negative.

Part C

1 point is earned for identifying $x = -4$ as a candidate

1 point is earned for $g(-4) = -1$

1 point is earned for justification and answer

The only x -value where f changes from negative to positive is $x = -4$. The other candidates for the location of the absolute minimum value are the endpoints.

$$g(-5) = 0$$

$$g(-4) = \int_{-3}^{-4} f(t) dt = -1$$

$$g(4) = \frac{9}{2} + \left(2 - \frac{\pi}{2}\right) = \frac{13-\pi}{2}$$

5-6b

So the absolute minimum value of g is -1.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for identifying $x = -4$ as a candidate

1 point is earned for $g(-4) = -1$

1 point is earned for justification and answer

The only x -value where f changes from negative to positive is $x = -4$. The other candidates for the location of the absolute minimum value are the endpoints.

$$g(-5) = 0$$

$$g(-4) = \int_{-3}^{-4} f(t) dt = -1$$

$$g(4) = \frac{9}{2} + \left(2 - \frac{\pi}{2}\right) = \frac{13 - \pi}{2}$$

So the absolute minimum value of g is -1.

Part D

Two points are earned for the correct values

<-1> each missing or extra value

$$x = -3, 1, 2$$



0	1	2
---	---	---

The student response earns all of the following points:

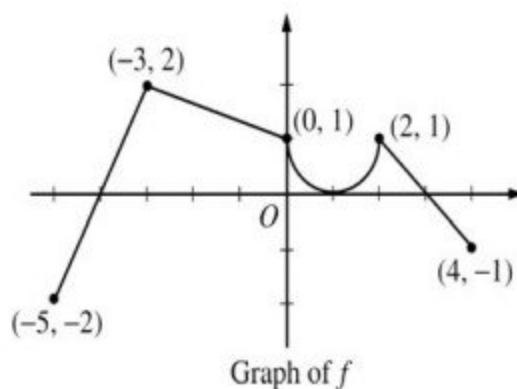
Two points are earned for the correct values

<-1> each missing or extra value

$$x = -3, 1, 2$$

5-6b

31.



The graph of the function f shown above consists of a semicircle and three line segments. Let g be the function given by $g(x) = \int_{-3}^x f(t) dt$.

Find all values of x in the open interval $(-5, 4)$ at which the graph of g has a point of inflection.

Part D

Two points are earned for the correct values

< -1 > each missing or extra value

$x = -3, 1, 2$



0	1	2
---	---	---

The student response earns all of the following points:

Two points are earned for the correct values

< -1 > each missing or extra value

$x = -3, 1, 2$

5-6b

32. Consider a differentiable function f having domain all positive real numbers, and for which it is known that $f'(x) = (4 - x)x^{-3}$ for $x > 0$.
- (a) Find the x -coordinate of the critical point of f . Determine whether the point is a relative maximum, a relative minimum, or neither for the function f . Justify your answer.
- (b) Find all intervals on which the graph of f is concave down. Justify your answer.
- (c) Given that $f(1) = 2$, determine the function f .

Part A

1 point is earned for $x = 4$

1 point is earned for relative maximum

1 point is earned for the justification

$$f'(x) = 0 \text{ at } x = 4$$

$$f'(x) > 0 \text{ for } 0 < x < 4$$

$$f'(x) < 0 \text{ for } x > 4$$

Therefore f has a relative maximum at $x = 4$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $x = 4$

1 point is earned for relative maximum

1 point is earned for the justification

$$f'(x) = 0 \text{ at } x = 4$$

$$f'(x) > 0 \text{ for } 0 < x < 4$$

$$f'(x) < 0 \text{ for } x > 4$$

Therefore f has a relative maximum at $x = 4$.

Part B

2 points are earned for $f''(x)$

5-6b

1 point is earned for answer with justification

$$\begin{aligned}
 f''(x) &= -x^{-3} + (4 - x)(-3x^{-4}) \\
 &= -x^{-3} - 12x^{-4} + 3x^{-3} \\
 &= 2x^{-4}(x - 6) \\
 &= \frac{2(x-6)}{x^4}
 \end{aligned}$$

$$f''(x) < 0 \text{ for } 0 < x < 6$$

The graph of f is concave down on the interval $0 < x < 6$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $f''(x)$

1 point is earned for answer with justification

$$\begin{aligned}
 f''(x) &= -x^{-3} + (4 - x)(-3x^{-4}) \\
 &= -x^{-3} - 12x^{-4} + 3x^{-3} \\
 &= 2x^{-4}(x - 6) \\
 &= \frac{2(x-6)}{x^4}
 \end{aligned}$$

$$f''(x) < 0 \text{ for } 0 < x < 6$$

The graph of f is concave down on the interval $0 < x < 6$.

Part C

1 point is earned for integral

1 point is earned for antiderivative

1 point is earned for the answer

5-6b

$$\begin{aligned}f(x) &= 2 + \int_1^x (4t^{-3} - t^{-2}) dt \\&= 2 + \left[-2t^{-2} + t^{-1}\right]_{t=1}^{t=x} \\&= 3 - 2x^{-2} + x^{-1}\end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for antiderivative

1 point is earned for the answer

$$\begin{aligned}f(x) &= 2 + \int_1^x (4t^{-3} - t^{-2}) dt \\&= 2 + \left[-2t^{-2} + t^{-1}\right]_{t=1}^{t=x} \\&= 3 - 2x^{-2} + x^{-1}\end{aligned}$$

5-6b

33. Consider a differentiable function f having domain all positive real numbers, and for which it is known that $f'(x) = (4-x)x^{-3}$ for $x > 0$.

Find all intervals on which the graph of f is concave down. Justify your answer.

Part B

2 points are earned for $f''(x)$

1 point is earned for answer with justification

$$\begin{aligned} f''(x) &= -x^3 + (4-x)(-3x^{-4}) \\ &= -x^{-3} - 12x^{-4} + 3x^{-3} \\ &= 2x^{-4}(x-6) \\ &= \frac{2(x-6)}{x^4} \end{aligned}$$

$$f''(x) < 0 \text{ for } 0 < x < 6$$

The graph of f is concave down on the interval $0 < x < 6$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $f''(x)$

1 point is earned for answer with justification

$$\begin{aligned} f''(x) &= -x^3 + (4-x)(-3x^{-4}) \\ &= -x^{-3} - 12x^{-4} + 3x^{-3} \\ &= 2x^{-4}(x-6) \\ &= \frac{2(x-6)}{x^4} \end{aligned}$$

$$f''(x) < 0 \text{ for } 0 < x < 6$$

The graph of f is concave down on the interval $0 < x < 6$.

5-6b

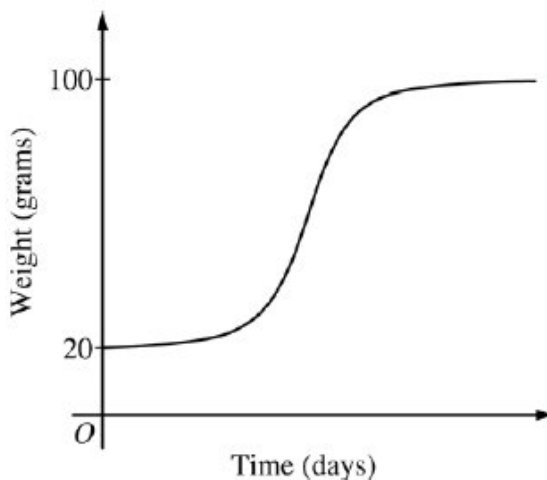
34. The rate at which a baby bird gains weight is proportional to the difference between its adult weight and its current weight. At time $t = 0$, when the bird is first weighed, its weight is 20 grams. If $B(t)$ is the weight of the bird, in grams, at time t days after it is first weighed, then

$$\frac{dB}{dt} = \frac{1}{5}(100 - B).$$

Let $y = B(t)$ be the solution to the differential equation above with initial condition $B(0) = 20$.

(a) Is the bird gaining weight faster when it weighs 40 grams or when it weighs 70 grams? Explain your reasoning.

(b) Find $\frac{d^2B}{dt^2}$ in terms of B . Use $\frac{d^2B}{dt^2}$ to explain why the graph of B cannot resemble the following graph.



(c) Use separation of variables to find $y = B(t)$, the particular solution to the differential equation with initial condition $B(0) = 20$.

Part A

1 point is earned for using $\frac{dB}{dt}$

1 point is earned for the answer with reason

$$\left. \frac{dB}{dt} \right|_{B=40} = \frac{1}{5}(60) = 12$$

$$\left. \frac{dB}{dt} \right|_{B=70} = \frac{1}{5}(30) = 6$$

Because $\left. \frac{dB}{dt} \right|_{B=40} > \left. \frac{dB}{dt} \right|_{B=70}$, the bird is gaining weight faster when it weighs 40 grams.



0	1	2
---	---	---

5-6b

The student response earns all of the following points:

1 point is earned for using $\frac{dB}{dt}$

1 point is earned for the answer with reason

$$\left. \frac{dB}{dt} \right|_{B=40} = \frac{1}{5}(60) = 12$$

$$\left. \frac{dB}{dt} \right|_{B=70} = \frac{1}{5}(30) = 6$$

Because $\left. \frac{dB}{dt} \right|_{B=40} > \left. \frac{dB}{dt} \right|_{B=70}$, the bird is gaining weight faster when it weighs 40 grams.

Part B

1 point is earned for the $\frac{d^2B}{dt^2}$ in terms of B

1 point is earned for the explanation

$$\frac{d^2B}{dt^2} = -\frac{1}{5} \frac{dB}{dt} = -\frac{1}{5} \cdot \frac{1}{5}(100 - B) = -\frac{1}{25}(100 - B)$$

Therefore, the graph of B is concave down for $20 \leq B < 100$. A portion of the given graph is concave up.



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for the $\frac{d^2B}{dt^2}$ in terms of B

1 point is earned for the explanation

$$\frac{d^2B}{dt^2} = -\frac{1}{5} \frac{dB}{dt} = -\frac{1}{5} \cdot \frac{1}{5}(100 - B) = -\frac{1}{25}(100 - B)$$

5-6b

Therefore, the graph of B is concave down for $20 \leq B < 100$. A portion of the given graph is concave up.

Part C

1 point is earned for the separation of variables

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for B

$$\frac{dB}{dt} = \frac{1}{5}(100 - B)$$

$$\int \frac{1}{100 - B} dB = \int \frac{1}{5} dt$$

$$-\ln|100 - B| = \frac{1}{5}t + c$$

Because $20 \leq B < 100$, $|100 - B| = 100 - B$.

$$-\ln(100 - 20) = \frac{1}{5}(0) + C \Rightarrow -\ln(80) = C$$

$$100 - B = 80e^{-t/5}$$

$$B(t) = 100 - 80e^{-t/5}, t \geq 0$$



0	1	2	3	4	5
---	---	---	---	---	---

The student earns all of the following points:

1 point is earned for the separation of variables

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for B

5-6b

$$\frac{dB}{dt} = \frac{1}{5}(100 - B)$$

$$\int \frac{1}{100 - B} dB = \int \frac{1}{5} dt$$

$$-\ln|100 - B| = \frac{1}{5}t + c$$

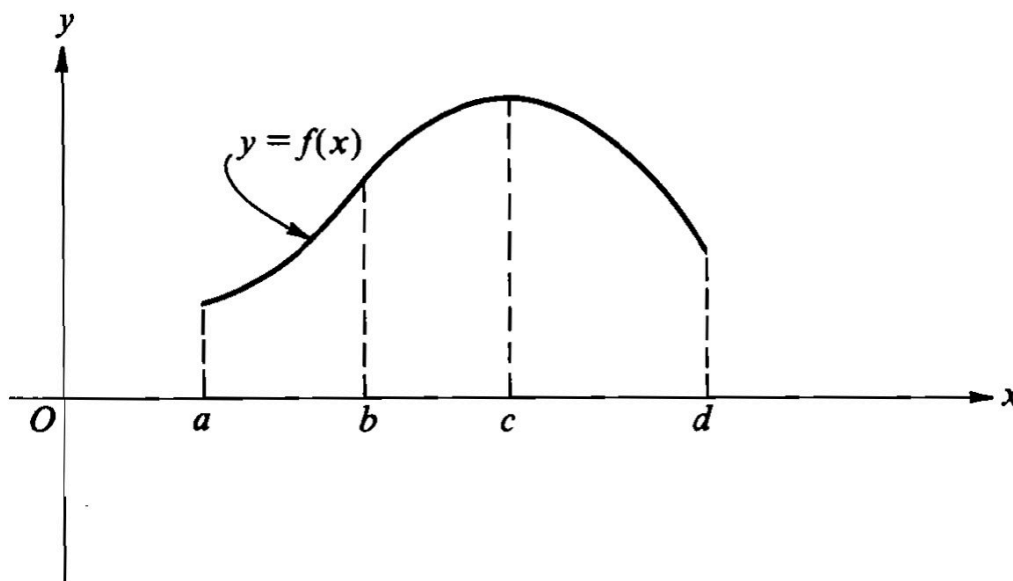
Because $20 \leq B < 100$, $|100 - B| = 100 - B$.

$$-\ln(100 - 20) = \frac{1}{5}(0) + C \Rightarrow -\ln(80) = C$$

$$100 - B = 80e^{-t/5}$$

$$B(t) = 100 - 80e^{-t/5}, t \geq 0$$

35.



The graph of $y = f(x)$ is shown in the figure above. On which of the following intervals are $\frac{dy}{dx} > 0$ and $\frac{d^2y}{dx^2} < 0$?

I. $a < x < b$ II. $b < x < c$ III. $c < x < d$

(A) I only

(B) II only

(C) III only

(D) I and II

(E) II and III



5-6b

36. Let f be a function defined by $f(x) = \begin{cases} 2x - x^2 & \text{for } x \leq 1, \\ x^2 + kx + p & \text{for } x > 1. \end{cases}$
- (a) For what values of k and p will f be continuous and differentiable at $x = 1$?
- (b) For the values of k and p found in part (a), on what interval or intervals is f increasing?
- (c) Using the values of k and p found in part (a), find all points of inflection of the graph of f . Support your conclusion.

Part A

1 point is earned for equation of $f(1)$

1 point is earned for $1 = 1 + k + p$

1 point is earned for $0 = 2 + k$

1 point is earned for solution of system

For continuity at $x = 1$,

$$\lim_{x \rightarrow 1^-} (2x - x^2) = f(1) = \lim_{x \rightarrow 1^+} (x^2 + kx + p)$$

Therefore $1 = 1 + k + p$

Since f is continuous at $x = 1$ and is piecewise polynomial, left and right derivatives exist.

$$f'_-(1) = 0 \text{ and } f'_+(1) = 2 + k$$

For differentiability at $x = 1$, $0 = 2 + k$.

Therefore $k = -2$, $p = 2$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for equation of $f(1)$

1 point is earned for $1 = 1 + k + p$

1 point is earned for $0 = 2 + k$

1 point is earned for solution of system

5-6b

For continuity at $x = 1$,

$$\lim_{x \rightarrow 1^-} (2x - x^2) = f(1) = \lim_{x \rightarrow 1^+} (x^2 + kx + p)$$

Therefore $1 = 1 + k + p$

Since f is continuous at $x = 1$ and is piecewise polynomial, left and right derivatives exist.

$$f'_-(1) = 0 \text{ and } f'_+(1) = 2 + k$$

For differentiability at $x = 1$, $0 = 2 + k$.

Therefore $k = -2$, $p = 2$.

Part B

1 point is earned for correct f' and analysis of sign of f' on domain

1 point is earned for $(-\infty, \infty)$

$$f'(x) = 2 - 2x, x \leq 1$$

$$2 - 2x > 0$$

$$x < 1$$

$$f'(x) = 2x - 2, x > 1$$

$$2x - 2 > 0$$

$$x > 1$$

Since f increases on each of $(-\infty, 1)$ and $(1, \infty)$ and is continuous at $x = 1$, f is increasing on $(-\infty, \infty)$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correct f' and analysis of sign of f' on domain

1 point is earned for $(-\infty, \infty)$

$$f'(x) = 2 - 2x, x \leq 1$$

5-6b

$$2 - 2x > 0$$

$$x < 1$$

$$f'(x) = 2x - 2, x > 1$$

$$2x - 2 > 0$$

$$x > 1$$

Since f increases on each of $(-\infty, 1)$ and $(1, \infty)$ and is continuous at $x = 1$, f is increasing on $(-\infty, \infty)$.

Part C

1 point is earned for f''

1 point is earned for noting sign changes in f'' or change of concavity in f

1 point is earned for ordered pair (or equivalent)

$$f''(x) = -2, x < 1$$

$$f''(x) = 2, x > 1$$

Since $f''(x) < 0$ on $(-\infty, 1)$ and

$f''(x) > 0$ on $(1, \infty)$ and

$f(1)$ is defined,

$(1, f(1)) = (1, 1)$ is a point of inflection.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for f''

1 point is earned for noting sign changes in f'' or change of concavity in f

1 point is earned for ordered pair (or equivalent)

$$f''(x) = -2, x < 1$$

$$f''(x) = 2, x > 1$$

Since $f''(x) < 0$ on $(-\infty, 1)$ and

5-6b

$f''(x) > 0$ on $(1, \infty)$ and

$f(1)$ is defined,

$(1, f(1)) = (1, 1)$ is a point of inflection.

37. Let f be a function defined by $f(x) = \begin{cases} 2x - x^2 & \text{for } x \leq 1, \\ x^2 + kx + p & \text{for } x > 1. \end{cases}$

Using the values of k and p found in part (a), find all points of inflection of the graph of f . Support your conclusion.

Part C

1 point is earned for f''

1 point is earned for noting sign changes in f'' or change of concavity in f

1 point is earned for ordered pair (or equivalent)

$$f''(x) = -2, x < 1$$

$$f''(x) = 2, x > 1$$

$$f''(x) < 0 \text{ on } (-\infty, 1) \text{ and}$$

$$f''(x) > 0 \text{ on } (1, \infty) \text{ and}$$

$f(1)$ is defined,

$(1, f(1)) = (1, 1)$ is a point of inflection.



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for f''

1 point is earned for noting sign changes in f'' or change of concavity in f

1 point is earned for ordered pair (or equivalent)

$$f''(x) = -2, x < 1$$

$$f''(x) = 2, x > 1$$

$$f''(x) < 0 \text{ on } (-\infty, 1) \text{ and}$$

$$f''(x) > 0 \text{ on } (1, \infty) \text{ and}$$

$f(1)$ is defined,

$(1, f(1)) = (1, 1)$ is a point of inflection.

5-6b

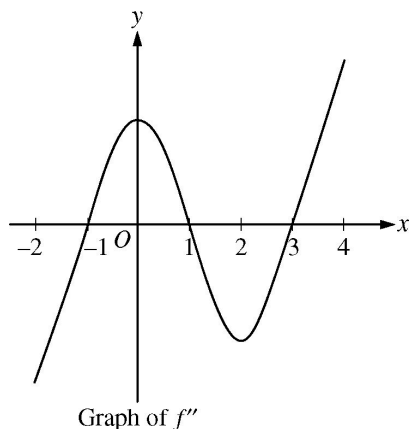
38.

x	0	1	2	3
$f''(x)$	5	0	-7	4

The polynomial function f has selected values of its second derivative f'' given in the table above. Which of the following statements must be true?

- (A) f is increasing on the interval $(0, 2)$.
- (B) f is decreasing on the interval $(0, 2)$.
- (C) f has a local maximum at $x = 1$
- (D) The graph of f has a point of inflection at $x = 1$
- (E) The graph of f changes concavity in the interval $(0, 2)$. ✓

39.



The graph of f'' , the second derivative of f , is shown above for $-2 \leq x \leq 4$. What are all intervals on which the graph of the function f is concave down?

- (A) $-1 < x < 1$
- (B) $0 < x < 2$
- (C) $1 < x < 3$ only
- (D) $-2 < x < -1$ only
- (E) $-2 < x < -1$ and $1 < x < 3$ ✓

40.  Let f be the function given by $f(x) = \cos(2x) + \ln(3x)$. What is the least value of x at which the graph of f changes concavity?

- (A) 0.56
- (B) 0.93 ✓
- (C) 1.18
- (D) 2.38
- (E) 2.44

5-6b

41. Let f be the function defined by $f(x) = 3x^5 - 5x^3 + 2$.
On what intervals is the graph of f concave upward?

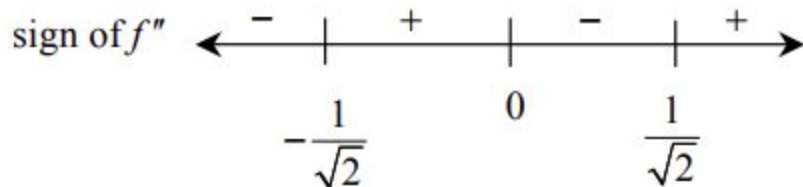
Part B

1 point is earned for $f''(x)$

1 point is earned for analyze sign of $f''(x)$ or explicitly setting students $f''(x) > 0$

1 point is earned for answer

$$f''(x) = 60x^3 - 30x = 30x(2x^2 - 1)$$



Answer: f is concave upward on $\left(-\frac{1}{\sqrt{2}}, 0\right)$ and on $\left(\frac{1}{\sqrt{2}}, \infty\right)$



0	1	2	3
---	---	---	---

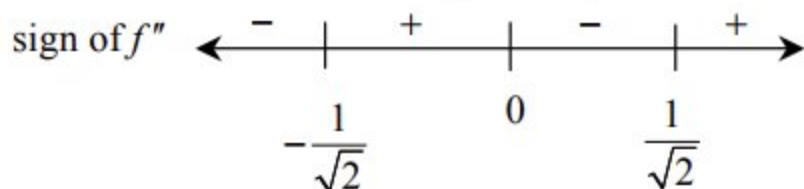
The student response earns three of the following points:

1 point is earned for $f''(x)$

1 point is earned for analyze sign of $f''(x)$ or explicitly setting students $f''(x) > 0$

1 point is earned for answer

$$f''(x) = 60x^3 - 30x = 30x(2x^2 - 1)$$



5-6b

Answer: f is concave upward on $\left(-\frac{1}{\sqrt{2}}, 0\right)$ and on $\left(\frac{1}{\sqrt{2}}, \infty\right)$

42. Let f be the function given by $f(x) = 2xe^x$. The graph of f is concave down when

(A) $x < -2$



(B) $x > -2$

(C) $x < -1$

(D) $x > -1$

(E) $x < 0$

5-6b

43. Consider the differential equation $\frac{dy}{dx} = \frac{1}{2}x + y - 1$

Find d^2y/dx^2 in terms of x and y . Describe the region in the xy -plane in which all solution curves to the differential equation are concave up.

Part B

2 points are earned for $\frac{d^2y}{dx^2}$

1 point is earned for the description

$$\frac{d^2y}{dx^2} = \frac{1}{2} + \frac{dy}{dx} = \frac{1}{2}x + y - \frac{1}{2}$$

Solution curves will be concave up on the half-plane above the line

$$y = -\frac{1}{2}x + \frac{1}{2}.$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $\frac{d^2y}{dx^2}$

1 point is earned for the description

$$\frac{d^2y}{dx^2} = \frac{1}{2} + \frac{dy}{dx} = \frac{1}{2}x + y - \frac{1}{2}$$

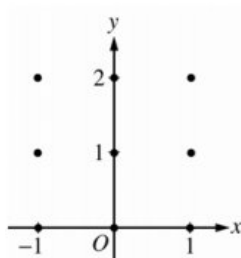
Solution curves will be concave up on the half-plane above the line

$$y = -\frac{1}{2}x + \frac{1}{2}.$$

5-6b

44. Consider the differential equation $\frac{dy}{dx} = \frac{1}{2}x + y - 1$.

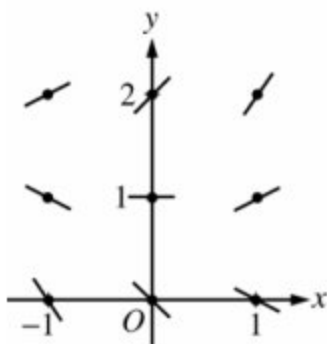
(a) On the axes provided, sketch a slope field for the given differential equation at the nine points indicated.



- (b) Find $\frac{d^2y}{dx^2}$ in terms of x and y . Describe the region in the xy -plane in which all solution curves to the differential equation are concave up.
- (c) Let $y = f(x)$ be a particular solution to the differential equation with the initial condition $f(0) = 1$. Does f have a relative minimum, a relative maximum, or neither at $x = 0$? Justify your answer.
- (d) Find the values of the constants m and b , for which $y = mx + b$ is a solution to the differential equation.

Part A

2 points are earned for sign of each slope at each point and relative steepness of slope lines in rows and columns



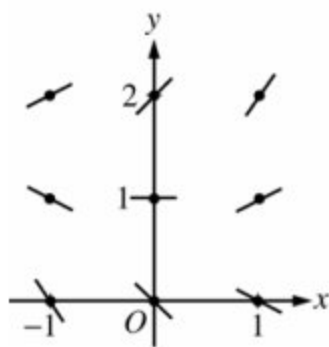
✓

0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for sign of each slope at each point and relative steepness of slope lines in rows and columns

5-6b

**Part B**

2 points are earned for $\frac{d^2y}{dx^2}$

1 point is earned for the description

$$\frac{d^2y}{dx^2} = \frac{1}{2} + \frac{dy}{dx} = \frac{1}{2}x + y - \frac{1}{2}$$

Solution curves will be concave up on the half-plane above the line

$$y = -\frac{1}{2}x + \frac{1}{2}.$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $\frac{d^2y}{dx^2}$

1 point is earned for the description

$$\frac{d^2y}{dx^2} = \frac{1}{2} + \frac{dy}{dx} = \frac{1}{2}x + y - \frac{1}{2}$$

Solution curves will be concave up on the half-plane above the line

$$y = -\frac{1}{2}x + \frac{1}{2}.$$

Part C

1 point is earned for the answer

5-6b

1 point is earned for the justification

$$\frac{dy}{dx}\bigg|_{(0,1)} = 0 + 1 - 1 = 0 \text{ and } \frac{d^2y}{dx^2}\bigg|_{(0,1)} = 0 + 1 - \frac{1}{2} > 0$$

Thus, f has a relative minimum at $(0,1)$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$$\frac{dy}{dx}\bigg|_{(0,1)} = 0 + 1 - 1 = 0 \text{ and } \frac{d^2y}{dx^2}\bigg|_{(0,1)} = 0 + 1 - \frac{1}{2} > 0$$

Thus, f has a relative minimum at $(0,1)$.

Part D

1 point is earned for the value of m

1 point is earned for the value of b

Substituting $y = mx + b$ into the differential equation:

$$m = \frac{1}{2}x + (mx + b) - 1 = (m + \frac{1}{2})x + (b - 1)$$

$$\text{Then } 0 = m + \frac{1}{2} \text{ and } m = b - 1: m = -\frac{1}{2} \text{ and } b = \frac{1}{2}.$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value of m

1 point is earned for the value of b

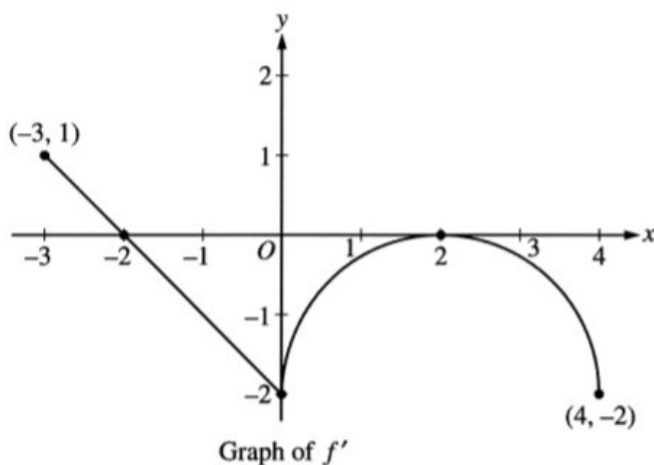
Substituting $y = mx + b$ into the differential equation:

5-6b

$$m = \frac{1}{2}x + (mx + b) - 1 = (m + \frac{1}{2})x + (b - 1)$$

Then $0 = m + \frac{1}{2}$ and $m = b - 1$: $m = -\frac{1}{2}$ and $b = \frac{1}{2}$.

45.



Let f be a function defined on the closed interval $-3 \leq x \leq 4$ with $f(0) = 3$. The graph of f' , the derivative of f , consists of one line segment and a semicircle, as shown above.

Find the x -coordinate of each point of inflection of the graph of f on the open interval $-3 < x < 4$. Justify your answer.

Part B

1 point is earned for the correct answers $x = 0$ and $x = 2$ only

1 point is earned for the justification

$x = 0$ and $x = 2$

f' changes from decreasing to increasing at $x = 0$ and from increasing to decreasing at $x = 2$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the correct answers $x = 0$ and $x = 2$ only

1 point is earned for the justification

$x = 0$ and $x = 2$

f' changes from decreasing to increasing at $x = 0$ and from increasing to decreasing at $x = 2$

5-6b

46. Consider the function f defined by $f(x) = e^x \cos x$ with domain $[0, 2\pi]$.
Find the x -coordinate of each point of inflection of the graph of f .

Part C

1 point is earned for $f''(x)$

1 point is earned for answer

$$\begin{aligned} f''(x) &= e^x [-\sin x - \cos x] + e^x [\cos x - \sin x] \\ &= -2e^x \sin x \end{aligned}$$

$$f''(x) = 0 \text{ when } x = 0, \pi, 2\pi$$

Point of inflection at $x = \pi$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for $f''(x)$

1 point is earned for answer

$$\begin{aligned} f''(x) &= e^x [-\sin x - \cos x] + e^x [\cos x - \sin x] \\ &= -2e^x \sin x \end{aligned}$$

$$f''(x) = 0 \text{ when } x = 0, \pi, 2\pi$$

Point of inflection at $x = \pi$

5-6b

47. Let f be the function defined by $f(x) = \ln(2 + \sin x)$ for $\pi \leq x \leq 2\pi$.

Find the x -coordinate of each inflection point on the graph of f . Justify your answer.

Part B

1 point is earned for $f''(x)$

2 point is earned for solution for $f''(x)=0$

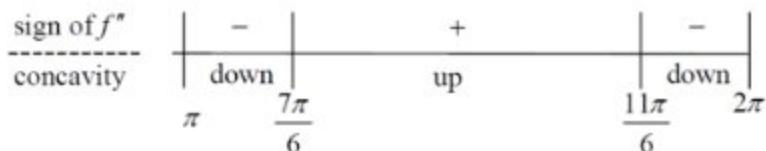
1 point is earned for specific justification

$$f''(x) = \frac{(-\sin x)(2 + \sin x) - \cos x \cos x}{(2 + \sin x)^2}$$

$$= \frac{-2 \sin x - 1}{(2 + \sin x)^2};$$

$$f''(x) = 0 \text{ when } \sin x = -\frac{1}{2}$$

$$x = \frac{7\pi}{6}, \frac{11\pi}{6}$$



$$x = \frac{7\pi}{6}, \frac{11\pi}{6} \text{ since concavity changes as indicated by these points}$$



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

1 point is earned for $f''(x)$

2 point is earned for solution for $f''(x)=0$

1 point is earned for specific justification

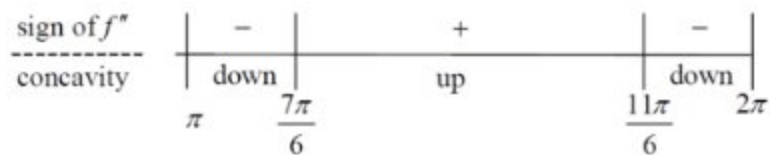
5-6b

$$f''(x) = \frac{(-\sin x)(2+\sin x) - \cos x \cos x}{(2+\sin x)^2}$$

$$= \frac{-2\sin x - 1}{(2+\sin x)^2},$$

$$f''(x) = 0 \text{ when } \sin x = -\frac{1}{2}$$

$$x = \frac{7\pi}{6}, \frac{11\pi}{6}$$



$$x = \frac{7\pi}{6}, \frac{11\pi}{6} \text{ since concavity changes as indicated by these points}$$

5-6b

48. Let f be the function defined by $f(x) = xe^{1-x}$ for all real numbers x .
Find the x -coordinate of each point of inflection of the graph of f .

Part C

1 point is earned for $f''(x)$ for student's $f'(x)$

1 point is earned for answer

$$\begin{aligned} f''(x) &= e^{1-x}(-1) + (1-x)e^{1-x}(-1) \\ &= (x-2)e^{1-x} \end{aligned}$$

Point of inflection at $x=2$.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for $f''(x)$ for student's $f'(x)$

1 point is earned for answer

$$\begin{aligned} f''(x) &= e^{1-x}(-1) + (1-x)e^{1-x}(-1) \\ &= (x-2)e^{1-x} \end{aligned}$$

Point of inflection at $x=2$.

49. The graph of $y = 5x^4 - x^5$ has a point of inflection at

(A) $(0, 0)$ only

(B) $(3, 162)$ only

(C) $(4, 256)$ only

(D) $(0, 0)$ and $(3, 162)$

(E) $(0, 0)$ and $(4, 256)$



Answer B

5-6b

50.  The derivative of the function f is given by $f'(x) = x^2 \cos(x^2)$. How many points of inflection does the graph of f have on the open interval $(-2, 2)$?
- (A) One
(B) Two
(C) Three
(D) Four
(E) Five ✓
51. An equation of the line tangent to $y = x^3 + 3x^2 + 2$ at its point of inflection is
- (A) $y = -6x - 6$
(B) $y = -3x + 1$ ✓
(C) $y = 2x + 10$
(D) $y = 3x - 1$
(E) $y = 4x + 1$
52. If f is the function defined by $f(x) = 3x^5 - 5x^4$, what are all the x -coordinates of points of inflection for the graph of f ?
- (A) -1
(B) 0
(C) 1 ✓
(D) 0 and 1
(E) -1, 0, and 1
53. At what value of x does the graph of $y = \frac{1}{x^2} - \frac{1}{x^3}$ have a point of inflection?
- (A) 0
(B) 1
(C) 2 ✓
(D) 3
(E) At no value of x
54. What is the x -coordinate of the point of inflection on the graph of $y = \frac{1}{3}x^3 + 5x^2 + 24$?
- (A) 5
(B) 0
(C) $-\frac{10}{3}$
(D) -5 ✓
(E) -10

5-6b

55. If $f''(x) = x(x+1)(x-2)^2$, then the graph of f has inflection points when $x =$
- (A) -1 only
 - (B) 2 only
 - (C) -1 and 0 only ✓
 - (D) -1 and 2 only
 - (E) -1, 0, and 2 only
56. If the graph of $y = x^3 + ax^2 + bx - 4$ has a point of inflection at $(1, -6)$, what is the value of b ?
- (A) -3
 - (B) 0 ✓
 - (C) 1
 - (D) 3
 - (E) It cannot be determined from the information given.
57. Let f be a function with a second derivative given by $f''(x) = x^2(x-3)(x-6)$. What are the x -coordinates of the points of inflection of the graph of f ?
- (A) 0 only
 - (B) 3 only
 - (C) 0 and 6 only
 - (D) 3 and 6 only ✓
 - (E) 0, 3, and 6

5-6b

58. Let f be the function defined by $f(x) = k\sqrt{x} - \ln x$, for $x > 0$, where k is a positive constant. For a certain value of the constant k , the graph of f has a point of inflection on the x -axis. Find this value of k .

Part C

1 point is earned for: $f'(x) = 0$ or $f(x) = 0$

1 point is earned for the equation in one variable

1 point is earned for the answer

At this inflection point, $f''(x) = 0$ and $f(x) = 0$.

$$f'(x) = 0 \Rightarrow \frac{-k}{4x^{3/2}} + \frac{1}{x^2} = 0 \Rightarrow k = \frac{4}{\sqrt{x}}$$

$$f(x) = 0 \Rightarrow k\sqrt{x} - \ln x = 0 \Rightarrow k = \frac{\ln x}{\sqrt{x}}$$

$$\text{Therefore, } \frac{4}{\sqrt{x}} = \frac{\ln x}{\sqrt{x}}$$

$$\Rightarrow 4 = \ln x$$

$$\Rightarrow x = e^4$$

$$\Rightarrow k = \frac{4}{e^2}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $f'(x) = 0$ or $f(x) = 0$

1 point is earned for the equation in one variable

1 point is earned for the answer

At this inflection point, $f''(x) = 0$ and $f(x) = 0$.

5-6b

$$f'(x) = 0 \Rightarrow \frac{-k}{4x^{3/2}} + \frac{1}{x^2} = 0 \Rightarrow k = \frac{4}{\sqrt{x}}$$

$$f(x) = 0 \Rightarrow k\sqrt{x} - \ln x = 0 \Rightarrow k = \frac{\ln x}{\sqrt{x}}$$

$$\text{Therefore, } \frac{4}{\sqrt{x}} = \frac{\ln x}{\sqrt{x}}$$

$$\Rightarrow 4 = \ln x$$

$$\Rightarrow x = e^4$$

$$\Rightarrow k = \frac{4}{e^2}$$

59.  Let f be the function with derivative defined by $f'(x) = \sin(x^3)$ on the interval $-1.8 < x < 1.8$. How many points of inflection does the graph of f have on this interval?

- (A) Two
(B) Three
(C) Four
(D) Five
(E) Six



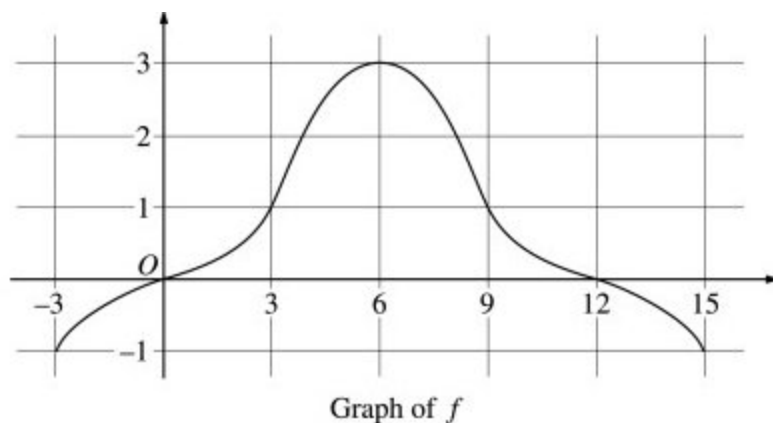
60.  The function f has first derivative given by $f'(x) = \frac{\sqrt{x}}{1+x+x^3}$. What is the x -coordinate of the inflection point of the graph of f ?

- (A) 1.008
(B) 0.473
(C) 0
(D) -0.278
(E) The graph of f has no inflection point.



5-6b

61.



The graph of a differentiable function f on the closed interval $[-3, 15]$ is shown in the figure above. The graph of f has a horizontal tangent line at $x = 6$. Let $g(x) = 5 + \int_6^x f(t) dt$ for $-3 \leq x \leq 15$.

On what intervals is the graph of g concave down? Justify your answer.

Part C

1 point is earned for the interval

1 point is earned for the justification

The graph of g is concave down on $(6, 15)$ since $g' = f$ is decreasing on this interval



0	1	2
---	---	---

The student response earns all of the following points:

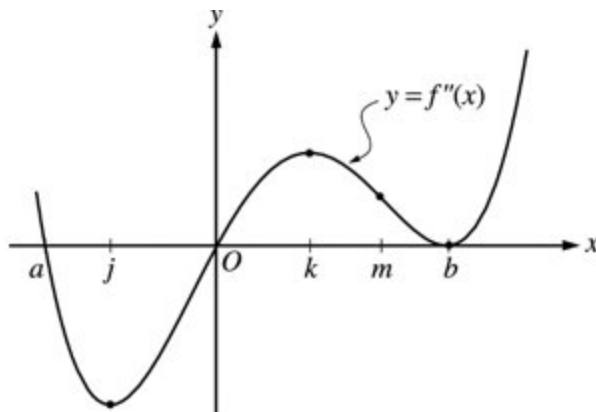
1 point is earned for the interval

1 point is earned for the justification

The graph of g is concave down on $(6, 15)$ since $g' = f$ is decreasing on this interval

5-6b

62.



The second derivative of the function f is given by $f''(x) = x(x - a)(x - b)^2$. The graph of f'' is shown above. For what values of x does the graph of f have a point of inflection?

(A) 0 and a only



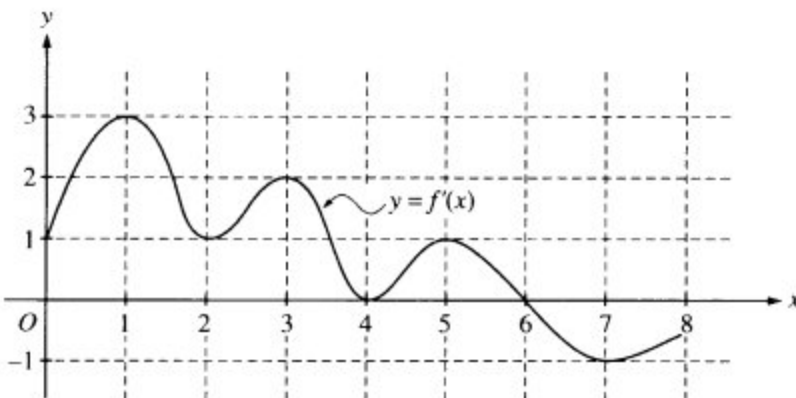
(B) 0 and m only

(C) b and j only

(D) 0, a , and b

(E) b , j , and k

63. Refer to the graph and the information below.



The function f is defined on the closed interval $[0, 8]$. The graph of its derivative f' is shown above.

How many points of inflection does the graph of f have?

(A) Two

(B) Three

(C) Four

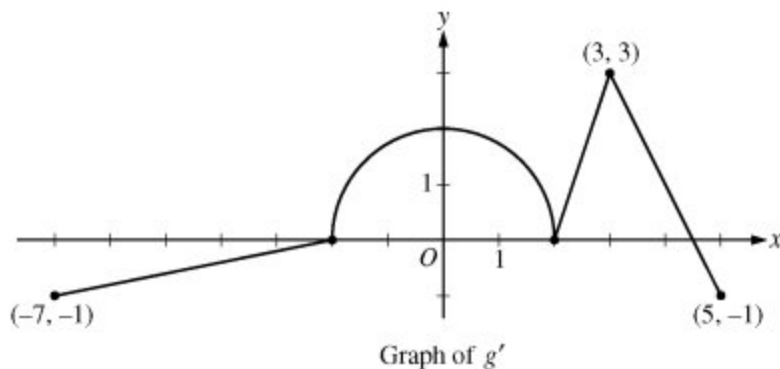
(D) Five

(E) Six



5-6b

64.



The function g is defined and differentiable on the closed interval $[-7, 5]$ and satisfies $g(0) = 5$. The graph of $y = g'(x)$, the derivative of g , consists of a semicircle and three line segments, as shown in the figure above.

Find the x -coordinate of each point of inflection of the graph of $y = g(x)$ on the interval $-7 < x < 5$. Explain your reasoning.

Part B

1 point is earned for identifying $x = 0, 2, 3$

1 point is earned for explanation

The graph of $y = g(x)$ has points of inflection at $x = 0$, $x = 2$, and $x = 3$ because g' changes from increasing to decreasing at $x = 0$ and $x = 3$, and g' changes from decreasing to increasing at $x = 2$.



0	1	2
---	---	---

The student response earns all of the following points:

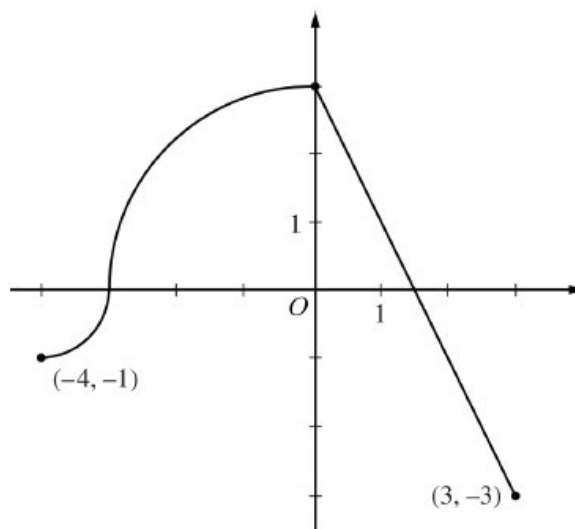
1 point is earned for identifying $x = 0, 2, 3$

1 point is earned for explanation

The graph of $y = g(x)$ has points of inflection at $x = 0$, $x = 2$, and $x = 3$ because g' changes from increasing to decreasing at $x = 0$ and $x = 3$, and g' changes from decreasing to increasing at $x = 2$.

5-6b

65.

Graph of f

The continuous function f is defined on the interval $-4 \leq x \leq 3$. The graph of f consists of two quarter circles and one line segment, as shown in the figure above. Let $g(x) = 2x + \int_0^x f(t) dt$.

Find all values of x on the interval $-4 < x < 3$ for which the graph of g has a point of inflection. Give a reason for your answer.

Part C

1 point is earned for answer with reason

$g''(x) = f(x)$ changes sign only at $x=0$. Thus the graph of g has a point of inflection at $x=0$.



0	1
---	---

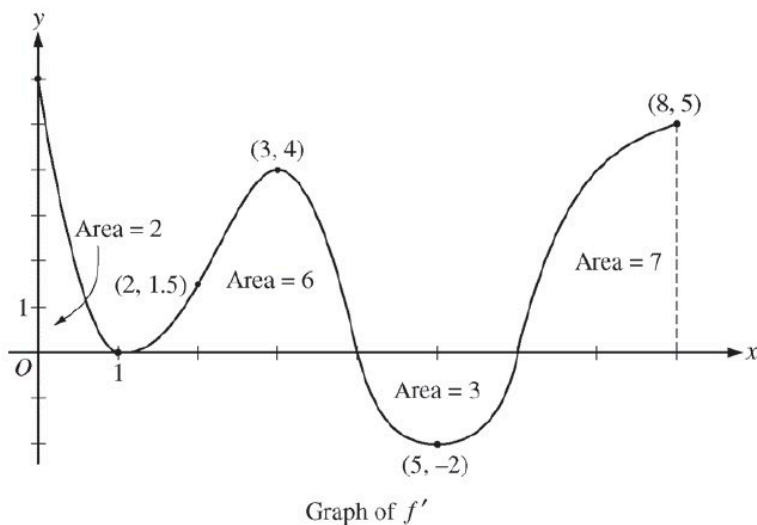
The student response earns all of the following points:

1 point is earned for answer with reason

$g''(x) = f(x)$ changes sign only at $x=0$. Thus the graph of g has a point of inflection at $x=0$.

5-6b

66.



The figure above shows the graph of f' the derivative of a twice-differentiable function f , on the closed interval $0 \leq x \leq 8$. The graph of f' has horizontal tangent lines at $x=1$, $x=3$, and $x=5$. The areas of the regions between the graph of f' and the x -axis are labeled in the figure. The function f is defined for all real numbers and satisfied $f(8)=4$.

On what open intervals contained in $0 < x < 8$ is the graph of f both concave down and increasing? Explain your reasoning.

Part C

1 point is earned for the answer

1 point is earned for the explanation

The graph of f is concave down and increasing on $0 < x < 1$ and $3 < x < 4$, because f' is decreasing and positive on these intervals.



0	1	2
---	---	---

The student response earns all of the following points:

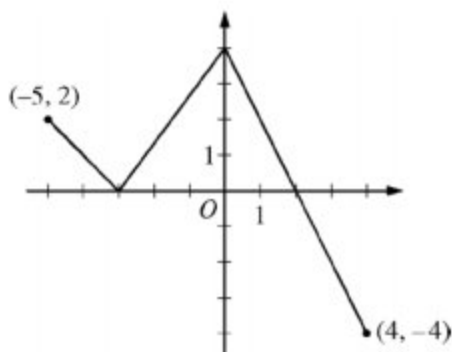
1 point is earned for the answer

1 point is earned for the explanation

The graph of f is concave down and increasing on $0 < x < 1$ and $3 < x < 4$, because f' is decreasing and positive on these intervals.

5-6b

67.

Graph of f

The function f is defined on the closed interval $[-5, 4]$. The graph of f consists of three line segments and is shown in the figure above. Let g be the function defined by $g(x) = \int_{-3}^x f(t) dt$.

On what open intervals contained in $-5 < x < 4$ is the graph of g both increasing and concave down? Give a reason for your answer.

Part B

1 point is earned for the answer

1 point is earned for the reason

$g'(x) = f(x)$ The graph of g is increasing and concave down on the intervals $-5 < x < -3$ and $0 < x < 2$ because $g' = f$ is positive and decreasing on these intervals.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

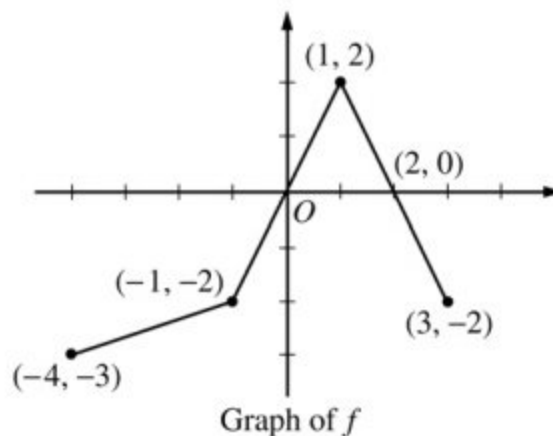
1 point is earned for the reason

$g'(x) = f(x)$

The graph of g is increasing and concave down on the intervals $-5 < x < -3$ and $0 < x < 2$ because $g' = f$ is positive and decreasing on these intervals.

5-6b

68.



The graph of the function f above consists of three line segments.

For the function g defined in part (a), find the x -coordinate of each point of inflection of the graph of g on the open interval $-4 < x < 3$. Explain your reasoning.

Part B

1 point is earned for $x=1$ (only)

1 point is earned for the reason

$x=1$

$g'=f$ changes from increasing to decreasing

at $x=1$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $x=1$ (only)

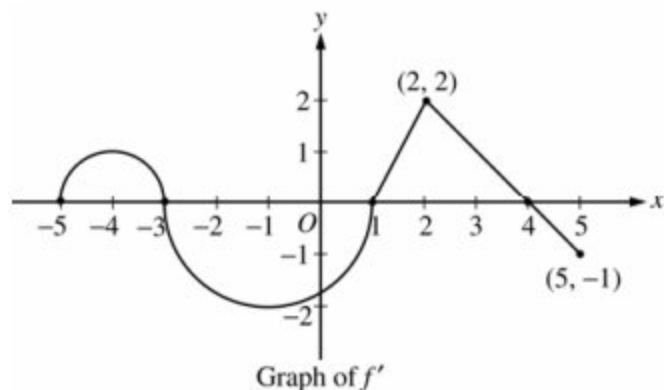
1 point is earned for the reason

$x=1$

$g'=f$ changes from increasing to decreasing

at $x=1$

5-6b



Let f be a function defined on the closed interval $-5 \leq x \leq 5$ with $f(1)=3$. The graph of f' , the derivative of f , consists of two semicircles and two line segments, as shown above.

69. For $-5 < x < 5$ find all values x at which the graph of f has a point of inflection. Justify your answer.

Part B

1 point is earned for the x-values

1 point is earned for the justification

f changes from increasing to decreasing, or vice versa, at

$x=-4$, -1 , and 2 . Thus, the graph of f has points of inflection when $x=-4$, -1 , and 2 .



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the x-values

1 point is earned for the justification

f changes from increasing to decreasing, or vice versa, at

$x=-4$, -1 , and 2 . Thus, the graph of f has points of inflection when $x=-4$, -1 , and 2 .

5-6b

70. Find all intervals on which the graph of f is concave up and also has positive slope. Explain your reasoning.
-

Part C

1 point is earned if considers intervals

1 point is earned for value and explanation

The graph of f is concave up with positive slope where f' is increasing and positive: $-5 < x < -4$ and $1 < x < 2$.



0	1	2
---	---	---

The student response earns two of the following points:

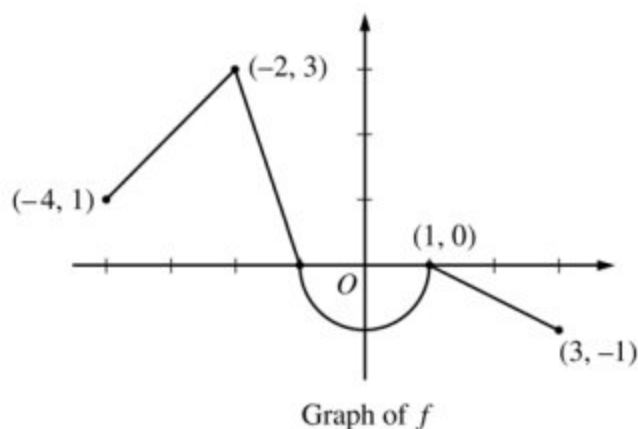
1 point is earned if considers intervals

1 point is earned for value and explanation

The graph of f is concave up with positive slope where f' is increasing and positive: $-5 < x < -4$ and $1 < x < 2$.

5-6b

71.



Let f be the continuous function defined on $[-4, 3]$ whose graph, consisting of three line segments and a semicircle centered at the origin, is given above. Let g be the function given by $g(x) = \int_1^x f(t) dt$.

For $-4 < x < 3$, find all values of x for which the graph of g has a point of inflection. Explain your reasoning.

Part D

1 point is earned for the answer

1 point is earned for the explanation

The graph of g has a point of inflection at each of $x=-2$, $x=0$, and $x=1$ because $g''(x)=f(x)$ changes sign at each of these values.



0	1	2
---	---	---

The student response earns all of the following points:

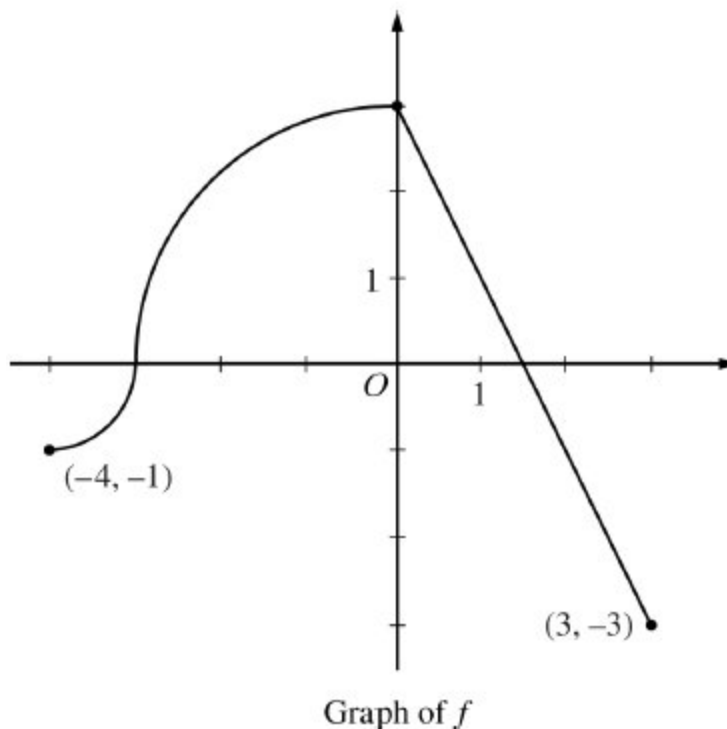
1 point is earned for the answer

1 point is earned for the explanation

The graph of g has a point of inflection at each of $x=-2$, $x=0$, and $x=1$ because $g''(x)=f(x)$ changes sign at each of these values.

5-6b

72.



The continuous function f is defined on the interval $-4 \leq x \leq 3$. The graph of f consists of two quarter circles and one line segment, as shown in the figure above. Let $g(x) = 2x + \int_0^x f(t) dt$.

- Find $g(-3)$. Find $g'(x)$ and evaluate $g'(-3)$.
- Determine the x -coordinate of the point at which g has an absolute maximum on the interval $-4 \leq x \leq 3$. Justify your answer.
- Find all values of x on the interval $-4 < x < 3$ for which the graph of g has a point of inflection. Give a reason for your answer.
- Find the average rate of change of f on the interval $-4 \leq x \leq 3$. There is no point c , $-4 < c < 3$, for which $f'(c)$ is equal to that average rate of change. Explain why this statement does not contradict the Mean Value Theorem.

Part A

1 point is earned for $g(-3)$

1 point is earned for $g'(x)$

1 point is earned for $g'(-3)$

$$g(-3) = 2(-3) + \int_0^{-3} f(t) dt = -6 - \frac{9\pi}{4}$$

$$g'(x) = 2 + f(x)$$

5-6b

$$g'(-3) = 2 + f(-3) = 2$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g(-3)$

1 point is earned for $g'(x)$

1 point is earned for $g'(-3)$

$$g(-3) = 2(-3) + \int_0^{-3} f(t)dt = -6 - \frac{9\pi}{4}$$

$$g'(x) = 2 + f(x)$$

$$g'(-3) = 2 + f(-3) = 2$$

Part B

1 point is earned for considering $g'(x) = 0$

1 point is earned for identifying interior candidate

1 point is earned for answer with justification

$g'(x) = 0$ when $f(x) = -2$. This occurs at $x = \frac{5}{2}$.

$g'(x) > 0$ for $-4 < x < \frac{5}{2}$ and $g'(x) < 0$ for $\frac{5}{2} < x < 3$.

Therefore g has an absolute maximum at $x = \frac{5}{2}$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $g'(x) = 0$

1 point is earned for identifying interior candidate

1 point is earned for answer with justification

5-6b

$g'(x) = 0$ when $f(x) = -2$. This occurs at $x = \frac{5}{2}$.

$g'(x) > 0$ for $-4 < x < \frac{5}{2}$ and $g'(x) < 0$ for $\frac{5}{2} < x < 3$.

Therefore g has an absolute maximum at $x = \frac{5}{2}$.

Part C

1 point is earned for answer with reason

$g''(x) = f'(x)$ changes sign only at $x = 0$. Thus the graph of g has a point of inflection at $x = 0$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for answer with reason

$g''(x) = f'(x)$ changes sign only at $x = 0$. Thus the graph of g has a point of inflection at $x = 0$.

Part D

1 point is earned for average rate of change

1 point is earned for explanation

The average rate of change of f on the interval $-4 \leq x \leq 3$ is $\frac{f(3) - f(-4)}{3 - (-4)} = -\frac{2}{7}$.

To apply the Mean Value Theorem, f must be differentiable at each point in the interval $-4 < x < 3$. However, f is not differentiable at $x = -3$ and $x = 0$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for average rate of change

1 point is earned for explanation

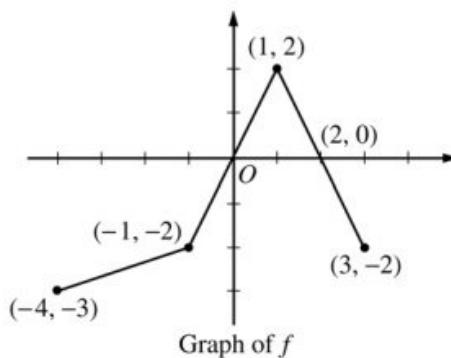
5-6b

The average rate of change of f on the interval $-4 \leq x \leq 3$ is $\frac{f(3) - f(-4)}{3 - (-4)} = -\frac{2}{7}$.

To apply the Mean Value Theorem, f must be differentiable at each point in the interval $-4 < x < 3$. However, f is not differentiable at $x = -3$ and $x = 0$.

5-6b

73.



The graph of the function f above consists of three line segments.

- (a) Let g be the function given by $g(x) = \int_{-4}^x f(t)dt$. For each of $g(-1)$, $g'(-1)$, and $g''(-1)$, find the value or state that it does not exist.
- (b) For the function g defined in part (a), find the x -coordinate of each point of inflection of the graph of g on the open interval $-4 < x < 3$. Explain your reasoning.
- (c) Let h be the function given by $h(x) = \int_x^3 f(t)dt$. Find all values of x in the closed interval $-4 \leq x \leq 3$ for which $h(x) = 0$.
- (d) For the function h defined in part (c), find all intervals on which h is decreasing. Explain your reasoning.

Part A

1 point is earned for $g(-1)$

1 point is earned for $g'(-1)$

1 point is earned for $g''(-1)$

$$g(-1) = \int_{-4}^{-1} f(t)dt = -\frac{1}{2}(3)(5) = -\frac{15}{2}$$

$$g'(-1) = f(-1) = -2$$

$g''(-1)$ does not exist because f is not differentiable at $x = -1$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g(-1)$

5-6b

1 point is earned for $g'(-1)$

1 point is earned for $g''(-1)$

$$g(-1) = \int_{-4}^{-1} f(t) dt = -\frac{1}{2}(3)(5) = -\frac{15}{2}$$

$$g'(-1) = f(-1) = -2$$

$g''(-1)$ does not exist because f is not differentiable at $x = -1$.

Part B

1 point is earned for $x = 1$ (only)

1 point is earned for the reason

$$x = 1$$

$g' = f$ changes from increasing to decreasing at $x = 1$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $x = 1$ (only)

1 point is earned for the reason

$$x = 1$$

$g' = f$ changes from increasing to decreasing at $x = 1$.

Part C

2 points are earned for the correct values

<-1> each missing or extra value

$$x = -1, 1, 3$$



0	1	2
---	---	---

5-6b

The student response earns all of the following points:

2 points are earned for the correct values

<-1> each missing or extra value

$$x = -1, 1, 3$$

Part D

1 point is earned for interval

1 point is earned for reason

h is decreasing on $[0, 2]$

$$h' = -f < 0 \text{ when } f > 0$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for interval

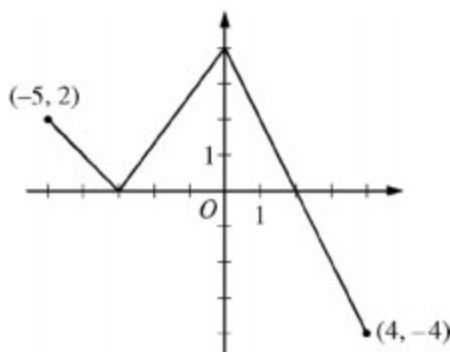
1 point is earned for reason

h is decreasing on $[0, 2]$

$$h' = -f < 0 \text{ when } f > 0$$

5-6b

74.

Graph of f

The function f is defined on the closed interval $[-5, 4]$. The graph of f consists of three line segments and is shown in the figure above. Let g be the function defined by $g(x) = \int_{-3}^x f(t) dt$.

(a) Find $g(3)$.

(b) On what open intervals contained in $-5 < x < 4$ is the graph of g both increasing and concave down? Give a reason for your answer.

(c) The function h is defined by $h(x) = \frac{g(x)}{5x}$. Find $h'(3)$.

(d) The function p is defined by $p(x) = f(x^2 - x)$. Find the slope of the line tangent to the graph of p at the point where $x = -1$.

Part A

1 point is earned for the answer

$$g(3) = \int_{-3}^3 f(t) dt = 6 + 4 - 1 = 9$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$g(3) = \int_{-3}^3 f(t) dt = 6 + 4 - 1 = 9$$

5-6b

Part B

1 point is earned for the answer

1 point is earned for the reason

$$g'(x) = f(x)$$

The graph of g is increasing and concave down on the intervals $-5 < x < -3$ and $0 < x < 2$ because $g' = f$ is positive and decreasing on these intervals.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the reason

$$g'(x) = f(x)$$

The graph of g is increasing and concave down on the intervals $-5 < x < -3$ and $0 < x < 2$ because $g' = f$ is positive and decreasing on these intervals.

Part C

2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = \frac{5xg'(x) - g(x)5}{(5x)^2} = \frac{5xg'(x) - 5g(x)}{25x^2}$$

$$\begin{aligned} h'(3) &= \frac{(5)(3)g'(3) - 5g(3)}{25 \cdot 3^2} \\ &= \frac{15(-2) - 5(9)}{225} = \frac{-75}{225} = -\frac{1}{3} \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

5-6b

2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = \frac{5xg'(x) - g(x)5}{(5x)^2} = \frac{5xg'(x) - 5g(x)}{25x^2}$$

$$h'(3) = \frac{(5)(3)g'(3) - 5g(3)}{25 \cdot 3^2}$$

$$= \frac{15(-2) - 5(9)}{225} = \frac{-75}{225} = -\frac{1}{3}$$

Part D

2 points are earned for $p'(x)$

1 point is earned for the answer

$$p'(x) = f'(x^2 - x)(2x - 1)$$

$$p'(-1) = f'(2)(-3) = (-2)(-3) = 6$$



0	1	2	3
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The student response earns all of the following points:

2 points are earned for $p'(x)$

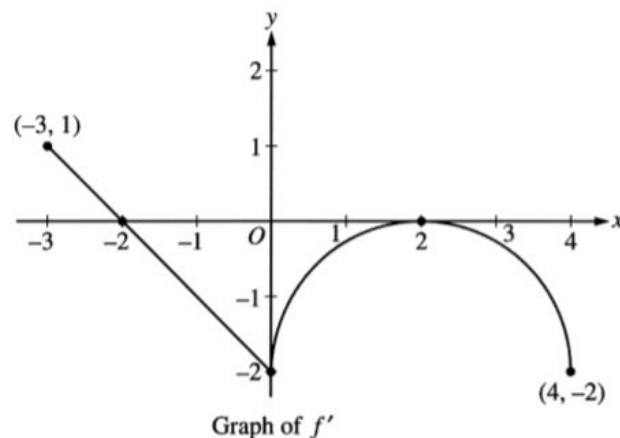
1 point is earned for the answer

$$p'(x) = f'(x^2 - x)(2x - 1)$$

$$p'(-1) = f'(2)(-3) = (-2)(-3) = 6$$

5-6b

75.



Let f be a function defined on the closed interval $-3 \leq x \leq 4$ with $f(0) = 3$. The graph of f' , the derivative of f , consists of one line segment and a semicircle, as shown above.

- On what intervals, if any, is f increasing? Justify your answer.
- Find the x -coordinate of each point of inflection of the graph of f on the open interval $-3 < x < 4$. Justify your answer.
- Find an equation for the line tangent to the graph of f at the point $(0, 3)$.
- Find $f(-3)$ and $f(4)$. Show the work that leads to your answers.

Part A

1 point is earned for the correct interval

1 point is earned for the correct reason

The function f is increasing on $[-3, -2]$ since $f' > 0$ for $-3 \leq x < -2$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct interval

1 point is earned for the correct reason

The function f is increasing on $[-3, -2]$ since $f' > 0$ for $-3 \leq x < -2$.

Part B

1 point is earned for the correct answers $x = 0$ and $x = 2$ only

1 point is earned for the justification

5-6b

$$x = 0 \text{ and } x = 2$$

f' changes from decreasing to increasing at $x = 0$ and from increasing to decreasing at $x = 2$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct answers $x = 0$ and $x = 2$ only

1 point is earned for the justification

$$x = 0 \text{ and } x = 2$$

f' changes from decreasing to increasing at $x = 0$ and from increasing to decreasing at $x = 2$

Part C

1 point is earned for the correct equation

$$f'(0) = -2$$

Tangent line is $y = -2x + 3$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for the correct equation

$$f'(0) = -2$$

Tangent line is $y = -2x + 3$.

Part D

1 point is earned for the difference of areas of triangles, $\pm \left(\frac{1}{2} - 2 \right)$

5-6b

$$f(0) - f(-3) = \int_{-3}^0 f'(t) dt$$

$$= \frac{1}{2}(1)(1) - \frac{1}{2}(2)(2) = -\frac{3}{2}$$

1 point is earned for the correct answer for $f(-3)$ using FTC

$$f(-3) = f(0) + \frac{3}{2} = \frac{9}{2}$$

1 point is earned for the area of rectangle - area of semicircle, $\pm \left(8 - \frac{1}{2}(2)^2\pi\right)$

$$f(4) - f(0) = \int_0^4 f'(t) dt$$

$$= -\left(8 - \frac{1}{2}(2)^2\pi\right) = -8 + 2\pi$$

1 point is earned for the correct answer of $f(4)$ using FTC

$$f(4) = f(0) - 8 + 2\pi = -5 + 2\pi$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the difference of areas of triangles, $\pm \left(\frac{1}{2} - 2\right)$

$$f(0) - f(-3) = \int_{-3}^0 f'(t) dt$$

$$= \frac{1}{2}(1)(1) - \frac{1}{2}(2)(2) = -\frac{3}{2}$$

1 point is earned for the correct answer for $f(-3)$ using FTC

$$f(-3) = f(0) + \frac{3}{2} = \frac{9}{2}$$

1 point is earned for the area of rectangle - area of semicircle, $\pm \left(8 - \frac{1}{2}(2)^2\pi\right)$

5-6b

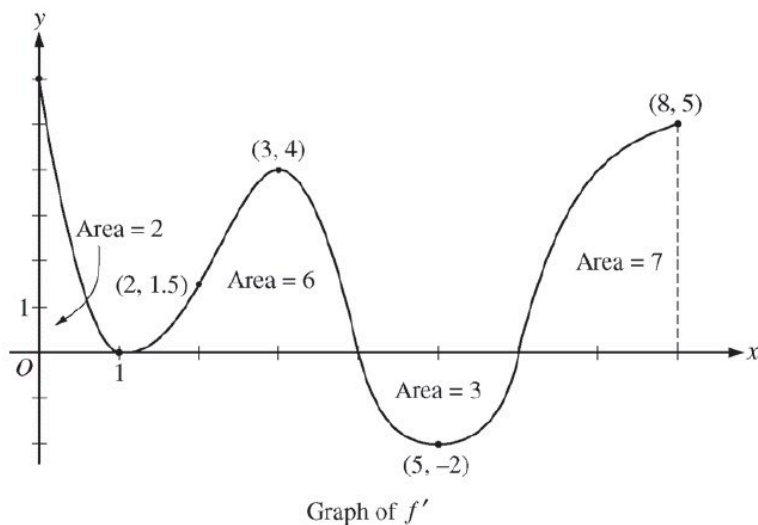
$$\begin{aligned} f(4) - f(0) &= \int_0^4 f'(t) dt \\ &= -\left(8 - \frac{1}{2}(2)^2\pi\right) = -8 + 2\pi \end{aligned}$$

1 point is earned for the correct answer of $f(4)$ using FTC

$$f(4) = f(0) - 8 + 2\pi = -5 + 2\pi$$

5-6b

76.



The figure above shows the graph of f' , the derivative of a twice-differentiable function f , on the closed interval $0 \leq x \leq 8$. The graph of f' has horizontal tangent lines at $x = 1$, $x = 3$, and $x = 5$. The areas of the regions between the graph of f' and the x -axis are labeled in the figure. The function f is defined for all real numbers and satisfies $f(8) = 4$.

- Find all values of x on the open interval $0 < x < 8$ for which the function f has a local minimum. Justify your answer.
- Determine the absolute minimum value of f on the closed interval $0 \leq x \leq 8$. Justify your answer.
- On what open intervals contained in $0 < x < 8$ is the graph of f both concave down and increasing? Explain your reasoning.
- The function g is defined by $g(x) = (f(x))^3$. If $f(3) = -5/2$, find the slope of the line tangent to the graph of g at $x = 3$.

Part A

1 point is earned for the answer with justification

$x = 6$ is the only critical point at which f' changes sign from negative to positive. Therefore, f has a local minimum at $x = 6$.



0

1

The student response earns all of the following points:

1 point is earned for the answer with justification

$x = 6$ is the only critical point at which f' changes sign from negative to positive. Therefore, f has a local minimum at $x = 6$.

5-6b

Part B

1 point is earned for considering $x = 0$ and $x = 6$

1 point is earned for the answer

1 point is earned for justification

From part (a), the absolute minimum occurs either at $x = 6$ or at an endpoint.

$$\begin{aligned} f(0) &= f(8) + \int_8^0 f'(x) dx \\ &= f(8) - \int_0^8 f'(x) dx = 4 - 12 = -8 \\ f(6) &= f(8) + \int_8^6 f'(x) dx \\ &= f(8) - \int_6^8 f'(x) dx = 4 - 7 = -3 \end{aligned}$$

$$f(8) = 4$$

The absolute minimum value of f on the closed interval $[0, 8]$ is -8.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $x = 0$ and $x = 6$

1 point is earned for the answer

1 point is earned for justification

From part (a), the absolute minimum occurs either at $x = 6$ or at an endpoint.

5-6b

$$f(0) = f(8) + \int_8^0 f'(x) dx$$

$$= f(8) - \int_0^8 f'(x) dx = 4 - 12 = -8$$

$$f(6) = f(8) + \int_8^6 f'(x) dx$$

$$= f(8) - \int_6^8 f'(x) dx = 4 - 7 = -3$$

$$f(8) = 4$$

The absolute minimum value of f on the closed interval $[0, 8]$ is -8.

Part C

1 point is earned for the answer

1 point is earned for the explanation

The graph of f is concave down and increasing on $0 < x < 1$ and $3 < x < 4$, because f' is decreasing and positive on these intervals.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the explanation

The graph of f is concave down and increasing on $0 < x < 1$ and $3 < x < 4$, because f' is decreasing and positive on these intervals.

Part D

2 points are earned for $g'(x)$

1 point is earned for the answer

5-6b

$$g'(x) = 3[f(x)]^2 \cdot f'(x)$$

$$g'(3) = 3[f(3)]^2 \cdot f'(3) = 3\left(-\frac{5}{2}\right)^2 \cdot 4 = 75$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $g'(x)$

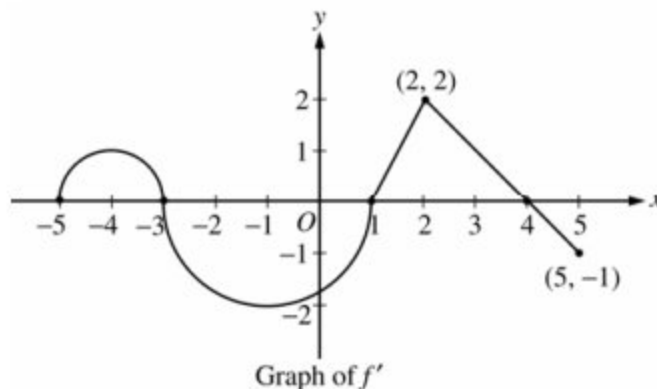
1 point is earned for the answer

$$g'(x) = 3[f(x)]^2 \cdot f'(x)$$

$$g'(3) = 3[f(3)]^2 \cdot f'(3) = 3\left(-\frac{5}{2}\right)^2 \cdot 4 = 75$$

5-6b

77.



Let f be a function defined on the closed interval $-5 \leq x \leq 5$ with $f(1) = 3$. The graph of f' , the derivative of f , consists of two semicircles and two line segments, as shown above.

- For $-5 < x < 5$, find all values x at which f has a relative maximum. Justify your answer.
- For $-5 < x < 5$, find all values x at which the graph of f has a point of inflection. Justify your answer.
- Find all intervals on which the graph of f is concave up and also has positive slope. Explain your reasoning.
- Find the absolute minimum value of $f(x)$ over the closed interval $-5 \leq x \leq 5$. Explain your reasoning.

Part A

1 point is earned for the x -values

1 point is earned for the justification

$$f'(x) = 0 \text{ at } x = -3, 1, 4$$

f' changes from positive to negative at -3 and 4 .

Thus, f has a relative maximum at $x = -3$ and at $x = 4$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the x -values

1 point is earned for the justification

$$f'(x) = 0 \text{ at } x = -3, 1, 4$$

5-6b

f' changes from positive to negative at $x = -3$ and $x = 4$.

Thus, f has a relative maximum at $x = -3$ and at $x = 4$.

Part B

1 point is earned for the x -values

1 point is earned for the justification

f' changes from increasing to decreasing, or vice versa, at $x = -4$, $x = -1$, and $x = 2$. Thus, the graph of f has points of inflection when $x = -4$, $x = -1$, and $x = 2$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the x -values

1 point is earned for the justification

f' changes from increasing to decreasing, or vice versa, at $x = -4$, $x = -1$, and $x = 2$. Thus, the graph of f has points of inflection when $x = -4$, $x = -1$, and $x = 2$.

Part C

1 point is earned if considers intervals

1 point is earned for value and explanation

The graph of f is concave up with positive slope where f' is increasing and positive: $-5 < x < -4$ and $1 < x < 2$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned if considers intervals

1 point is earned for value and explanation

The graph of f is concave up with positive slope where f' is increasing and positive: $-5 < x < -4$ and $1 < x < 2$.

5-6b

Part D

1 point is earned if identifies $x = 1$ as a candidate

1 point is earned for considering endpoint

1 point is earned for value and explanation

Candidates for the absolute minimum are where f' changes from negative to positive (at $x = 1$) and at the endpoints ($x = -5, 5$).

$$f(-5) = 3 + \int_1^{-5} f(x) dx = 3 - \frac{\pi}{2} + 2\pi > 3$$

$$f(1) = 3$$

$$f(5) = 3 + \int_1^{-5} f'(x) dx = 3 + \frac{3 \cdot 2}{2} - \frac{1}{2} > 3$$

The absolute minimum value of f on $[-5, 5]$ is $f(1) = 3$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if identifies $x = 1$ as a candidate

1 point is earned for considering endpoint

1 point is earned for value and explanation

Candidates for the absolute minimum are where f' changes from negative to positive (at $x = 1$) and at the endpoints ($x = -5, 5$).

$$f(-5) = 3 + \int_1^{-5} f(x) dx = 3 - \frac{\pi}{2} + 2\pi > 3$$

$$f(1) = 3$$

$$f(5) = 3 + \int_1^{-5} f'(x) dx = 3 + \frac{3 \cdot 2}{2} - \frac{1}{2} > 3$$

The absolute minimum value of f on $[-5, 5]$ is $f(1) = 3$.

5-6b

78. Let f be the function defined by $f(x) = 3x^5 - 5x^3 + 2$.

- On what intervals is f increasing?
- On what intervals is the graph of f concave upward?
- Write the equation of each horizontal tangent line to the graph of f .

Part A

1 point is earned for $f'(x)$

1 point is earned for analyze sign of $f'(x)$ or explicitly setting students $f'(x) > 0$

1 point is earned for answer

$$f'(x) = 15x^4 - 15x^2 = 15x^2(x^2 - 1)$$



Answer: f is increasing on the intervals $(-\infty, -1]$ and $[1, \infty)$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $f'(x)$

1 point is earned for analyze sign of $f'(x)$ or explicitly setting students $f'(x) > 0$

1 point is earned for answer

$$f'(x) = 15x^4 - 15x^2 = 15x^2(x^2 - 1)$$



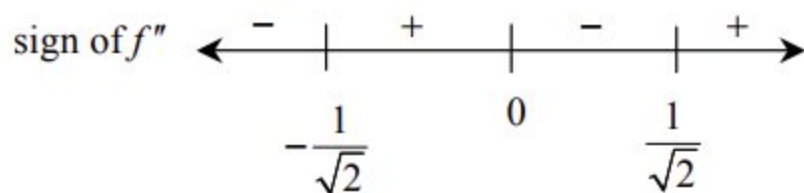
Answer: f is increasing on the intervals $(-\infty, -1]$ and $[1, \infty)$

5-6b

Part B1 point is earned for $f''(x)$ 1 point is earned for analyze sign of $f''(x)$ or explicitly setting students $f''(x) > 0$

1 point is earned for answer

$$f''(x) = 60x^3 - 30x = 30x(2x^2 - 1)$$



Answer: f is concave upward on $\left(-\frac{1}{\sqrt{2}}, 0\right)$ and on $\left(\frac{1}{\sqrt{2}}, \infty\right)$



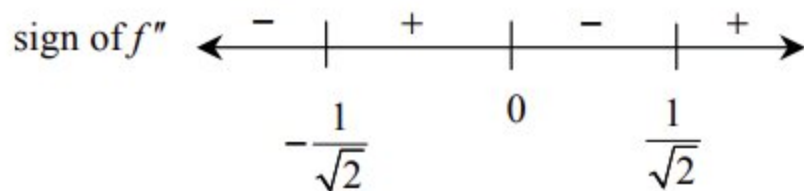
0	1	2	3
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The student response earns all of the following points:

1 point is earned for $f''(x)$ 1 point is earned for analyze sign of $f''(x)$ or explicitly setting students $f''(x) > 0$

1 point is earned for answer

$$f''(x) = 60x^3 - 30x = 30x(2x^2 - 1)$$



Answer: f is concave upward on $\left(-\frac{1}{\sqrt{2}}, 0\right)$ and on $\left(\frac{1}{\sqrt{2}}, \infty\right)$

Part C

5-6b

1 point is earned for solving student's $f'(x) = 0$

1 point is earned for answer, $y = \text{number}$

1 point is earned for consistent answer

$f'(x) = 0$ when $x = -1, 0, 1$

$x = -1 \Rightarrow f(x) = 4; y = 4$

$f(0) = 2; y = 2$

$f(1) = 0; y = 0$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for solving student's $f'(x) = 0$

1 point is earned for answer, $y = \text{number}$

1 point is earned for consistent answer

$f'(x) = 0$ when $x = -1, 0, 1$

$x = -1 \Rightarrow f(x) = 4; y = 4$

$f(0) = 2; y = 2$

$f(1) = 0; y = 0$

5-6b

79. Consider the function f defined by $f(x) = e^x \cos x$ with domain $[0, 2\pi]$.
- (a) Find the absolute maximum and minimum values of $f(x)$.
 - (b) Find the intervals on which f is increasing.
 - (c) Find the x -coordinate of each point of inflection of the graph of f .

Part A

1 point is earned for derivative

1 point is earned for finding critical values

1 point is earned for evaluating at critical points and ends

1 point is earned for choosing absolute max

1 point is earned for absolute min

$$\begin{aligned} f'(x) &= -e^x \sin x + e^x \cos x \\ &= e^x [\cos x - \sin x] \end{aligned}$$

$$f(x) = 0 \text{ when } \sin x = \cos x, x = \frac{\pi}{4}, \frac{5\pi}{4}$$

x	$f(x)$
0	1
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2} e^{\pi/4}$
$\frac{5\pi}{4}$	$-\frac{\sqrt{2}}{2} e^{5\pi/4}$
2π	$e^{2\pi}$

$$\text{Max: } e^{2\pi}; \text{ Min: } -\frac{\sqrt{2}}{2} e^{5\pi/4}$$



0	1	2	3	4	5
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The student response earns all of the following points:

5-6b

1 point is earned for derivative

1 point is earned for finding critical values

1 point is earned for evaluating at critical points and ends

1 point is earned for choosing absolute max

1 point is earned for absolute min

$$\begin{aligned} f'(x) &= -e^x \sin x + e^x \cos x \\ &= e^x [\cos x - \sin x] \end{aligned}$$

$$f'(x) = 0 \text{ when } \sin x = \cos x, x = \frac{\pi}{4}, \frac{5\pi}{4}$$

x	$f(x)$
0	1
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}e^{\pi/4}$
$\frac{5\pi}{4}$	$-\frac{\sqrt{2}}{2}e^{5\pi/4}$
2π	$e^{2\pi}$

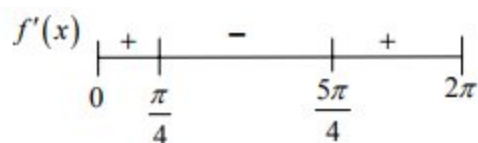
$$\text{Max: } e^{2\pi}; \text{ Min: } -\frac{\sqrt{2}}{2}e^{5\pi/4}$$

Part B

1 point is earned for positive intervals

1 point is earned for negative intervals

1 point is **deducted** for intervals outside of domain



Increasing on $[0, \frac{\pi}{4}]$, $[\frac{5\pi}{4}, 2\pi]$

5-6b



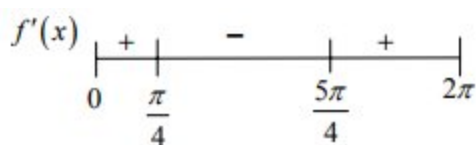
0	1	2
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The student response earns all of the following points:

1 point is earned for positive intervals

1 point is earned for negative intervals

1 point is **deducted** for intervals outside of domain



Increasing on $[0, \frac{\pi}{4}]$, $[\frac{5\pi}{4}, 2\pi]$

Part C

1 point is earned for $f''(x)$

1 point is earned for answer

$$\begin{aligned} f''(x) &= e^x [-\sin x - \cos x] + e^x [\cos x - \sin x] \\ &= -2e^x \sin x \end{aligned}$$

$$f''(x) = 0 \text{ when } x = 0, \pi, 2\pi$$

Point of inflection at $x = \pi$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $f''(x)$

1 point is earned for answer

$$\begin{aligned} f''(x) &= e^x [-\sin x - \cos x] + e^x [\cos x - \sin x] \\ &= -2e^x \sin x \end{aligned}$$

5-6b $f''(x) = 0$ when $x = 0, \pi, 2\pi$ Point of inflection at $x = \pi$

5-6b

80. Let f be the function defined by $f(x) = \ln(2 + \sin x)$ for $\pi \leq x \leq 2\pi$.

(a) Find the absolute maximum value and the absolute minimum value of f . Show the analysis that leads to your conclusion.

(b) Find the x -coordinate of each inflection point on the graph of f . Justify your answer.

Part A

1 point is earned for $f'(x)$

1 point is earned for solutions for $f'(x) = 0$ which includes at least one solution

1 point is earned for evaluation f at π and 2π

1 point is earned for evaluation of f at students critical point(s)

1 point is earned for answer

$$f'(x) = \frac{1}{2 + \sin x} \cos x ; \text{ In } [\pi, 2\pi], \cos x = 0 \text{ when } x = 3\pi/2;$$

x	$f(x)$
π	$\ln(2) = 0.693$
2π	$\ln(2)$
$\frac{3\pi}{2}$	$\ln(1) = 0$

absolute maximum value is $\ln 2$

absolute minimum value is 0



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $f'(x)$

1 point is earned for solutions for $f'(x) = 0$ which includes at least one solution

1 point is earned for evaluation f at π and 2π

5-6b

1 point is earned for evaluation of f at students critical point(s)

1 point is earned for answer

$$f'(x) = \frac{1}{2 + \sin x} \cos x; \text{ In } [\pi, 2\pi], \cos x = 0 \text{ when } x = 3\pi/2;$$

x	$f(x)$
π	$\ln(2) = 0.693$
2π	$\ln(2)$
$\frac{3\pi}{2}$	$\ln(1) = 0$

absolute maximum value is $\ln 2$

absolute minimum value is 0

Part B

1 point is earned for $f''(x)$

2 points are earned for solution for $f''(x) = 0$

1 point is earned for specific justification

$$\begin{aligned} f''(x) &= \frac{(-\sin x)(2 + \sin x) - \cos x \cos x}{(2 + \sin x)^2} \\ &= \frac{-2\sin x - 1}{(2 + \sin x)^2}; \end{aligned}$$

$$f''(x) = 0 \text{ when } \sin x = -1/2$$

$$x = 7\pi/6, 11\pi/6$$

sign of f''	-	+	-
concavity	down	up	down
	π	$\frac{7\pi}{6}$	$\frac{11\pi}{6}$
			2π

$x = 7\pi/6$ and $11\pi/6$ since concavity changes as indicated at these points

5-6b



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $f''(x)$

2 points are earned for solution for $f''(x) = 0$

1 point is earned for specific justification

$$f''(x) = \frac{(-\sin x)(2 + \sin x) - \cos x \cos x}{(2 + \sin x)^2}$$

$$= \frac{-2 \sin x - 1}{(2 + \sin x)^2};$$

$$f''(x) = 0 \text{ when } \sin x = -1/2$$

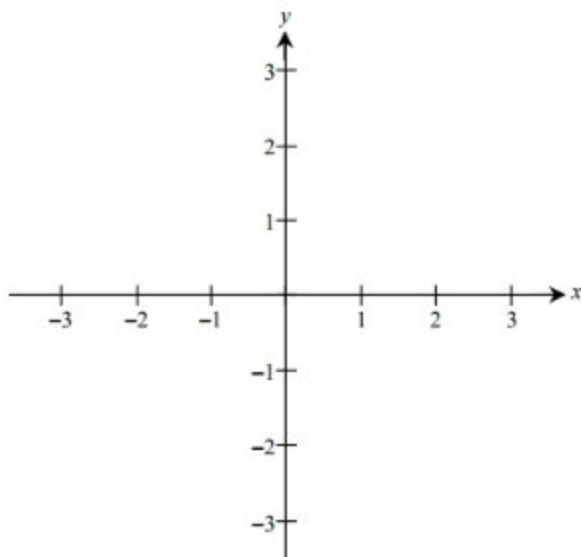
$$x = 7\pi/6, 11\pi/6$$

sign of f''	-	+	-
concavity	down	up	down
	π	$\frac{7\pi}{6}$	$\frac{11\pi}{6}$
			2π

$x = 7\pi/6$ and $11\pi/6$ since concavity changes as indicated at these points

5-6b

81. Let f be the function defined by $f(x) = xe^{1-x}$ for all real numbers x .
- Find each interval on which f is increasing.
 - Find the range of f .
 - Find the x -coordinate of each point of inflection of the graph of f .
 - Using the results found in parts (a), (b), and (c), sketch the graph of f in the xy -plane provided below. (Indicate all intercepts.)



Part A

1 point is earned for derivative

1 point is earned for answer

$$f'(x) = xe^{1-x}(-1) + e^{1-x} = (1-x)e^{1-x}$$

f increases on $(-\infty, 1]$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for derivative

1 point is earned for answer

5-6b

$$f'(x) = xe^{1-x}(-1) + e^{1-x} = (1-x)e^{1-x}$$

f increases on $(-\infty, 1]$

Part B

1 point is earned for left hand endpoint

1 point is earned for right hand endpoint

Note: 0/2 for All reals without supporting work.

$$f(1) = 1; \lim_{x \rightarrow -\infty} f(x) = -\infty$$

Range: $(-\infty, 1]$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for left hand endpoint

1 point is earned for right hand endpoint

Note: 0/2 for All reals without supporting work.

$$f(1) = 1; \lim_{x \rightarrow -\infty} f(x) = -\infty$$

Range: $(-\infty, 1]$

Part C

1 point is earned for $f''(x)$ for student's $f(x)$

1 point is earned for answer

$$\begin{aligned} f''(x) &= e^{1-x}(-1) + (1-x)e^{1-x}(-1) \\ &= (x-2)e^{1-x} \end{aligned}$$

Point of inflection at $x = 2$

5-6b

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $f''(x)$ for student's $f'(x)$

1 point is earned for answer

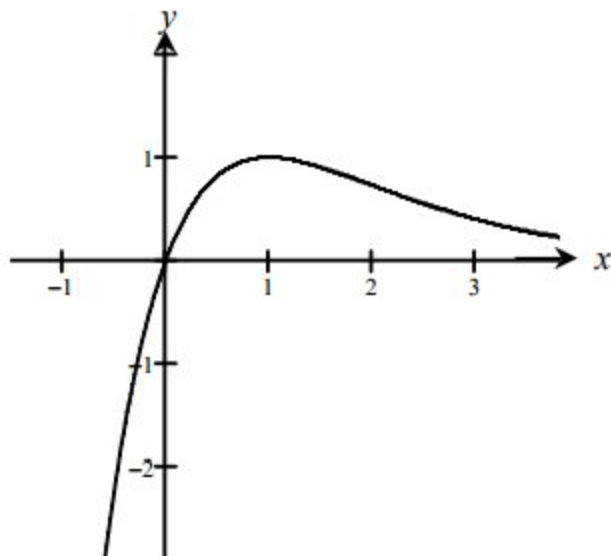
$$\begin{aligned} f''(x) &= e^{1-x}(-1) + (1-x)e^{1-x}(-1) \\ &= (x-2)e^{1-x} \end{aligned}$$

Point of inflection at $x = 2$

Part D

3 points are earned for answer without mistake

1 point is **deducted** for each error: intercept, domain, increasing, decreasing, range, inflection points.



✓

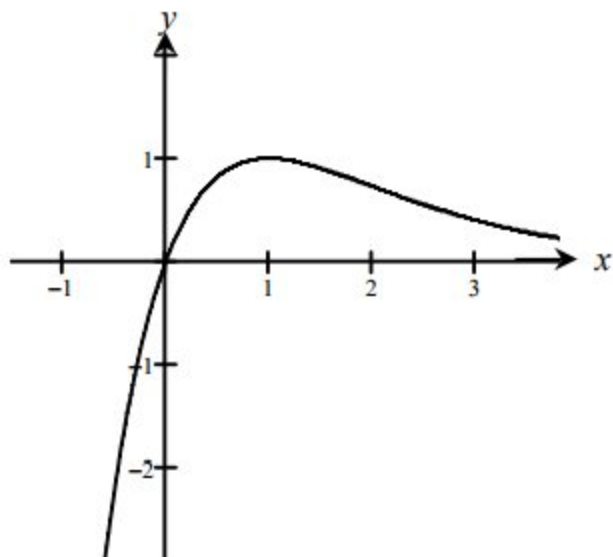
0	1	2	3
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The student response earns all of the following points:

3 points are earned for answer without mistake

5-6b

1 point is **deducted** for each error: intercept, domain, increasing, decreasing, range, inflection points.



5-6b

82. Let f be the function defined by $f(x) = k\sqrt{x} - \ln x$, for $x > 0$, where k is a positive constant.

(a) Find $f'(x)$ and $f''(x)$.

(b) For what value of the constant k does f have a critical point at $x=1$? For this value of k , determine whether f has a relative minimum, relative maximum, or neither at $x=1$. Justify your answer.

(c) For a certain value of the constant k , the graph of f has a point of inflection on the x -axis. Find this value of k .

Part A

1 point is earned for: $f'(x)$

1 point is earned for: $f''(x)$

$$f'(x) = \frac{k}{2\sqrt{x}} - \frac{1}{x}$$

$$f''(x) = -\frac{1}{4}kx^{-3/2} + x^{-2}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for: $f'(x)$

1 point is earned for: $f''(x)$

$$f'(x) = \frac{k}{2\sqrt{x}} - \frac{1}{x}$$

$$f''(x) = -\frac{1}{4}kx^{-3/2} + x^{-2}$$

Part B

1 point is earned if sets $f'(1) = 0$ or $f'(x) = 0$

1 point is earned if solves for k

1 point is earned for the answer

1 point is earned for the justification

5-6b

$$f'(1) = \frac{1}{2}k - 1 = 0 \Rightarrow k = 2$$

When $k = 2$, $f'(1) = 0$ and $f''(1) = -\frac{1}{2} + 1 > 0$.

f has a relative minimum value at $x = 1$ by the Second Derivative Test.



0	1	2	3	4
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f has a relative minimum value at $x = 1$ by the Second Derivative Test.

Part C

1 point is earned for: $f''(x) = 0$ or $f(x) = 0$

1 point is earned for the equation in one variable

1 point is earned for the answer

At this inflection point, $f''(x) = 0$ and $f(x) = 0$.

5-6b

$$f''(x) = 0 \Rightarrow \frac{-k}{4x^{3/2}} + \frac{1}{x^2} = 0 \Rightarrow k = \frac{4}{\sqrt{x}}$$

$$f(x) = 0 \Rightarrow k\sqrt{x} - \ln x = 0 \Rightarrow k = \frac{\ln x}{\sqrt{x}}$$

$$\text{Therefore, } \frac{4}{\sqrt{x}} = \frac{\ln x}{\sqrt{x}}$$

$$\Rightarrow 4 = \ln x$$

$$\Rightarrow x = e^4$$

$$\Rightarrow k = \frac{4}{e^2}$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for: $f''(x) = 0$ or $f(x) = 0$

1 point is earned for the equation in one variable

1 point is earned for the answer

At this inflection point, $f''(x) = 0$ and $f(x) = 0$.

$$f''(x) = 0 \Rightarrow \frac{-k}{4x^{3/2}} + \frac{1}{x^2} = 0 \Rightarrow k = \frac{4}{\sqrt{x}}$$

$$f(x) = 0 \Rightarrow k\sqrt{x} - \ln x = 0 \Rightarrow k = \frac{\ln x}{\sqrt{x}}$$

$$\text{Therefore, } \frac{4}{\sqrt{x}} = \frac{\ln x}{\sqrt{x}}$$

$$\Rightarrow 4 = \ln x$$

$$\Rightarrow x = e^4$$

$$\Rightarrow k = \frac{4}{e^2}$$